# Prealgebra Lessons

**Kelly Stady**

**July 11, 2026**

# Contents

# Part I

# Table of Contents

## 1 Exponents and Positive Powers of 10

### Exponential Notation

Exponential notation is shorthand for repeated multiplication of the same quantity. For example, the product $5 \cdot 5 \cdot 5$ can be rewritten more simply as $5^3$. In this example, the number 5 is called the **base** and the number 3 is called the **exponent**. In general, the exponent indicates the number of times to multiply the base.

**Problem 1** *Identify the base and exponent of* $10^7$.

***Base*** $= 10$ ***Exponent*** $= 7$

**Problem 2** *Identify the base and exponent of* $4^5$.

***Base*** $= 4$ ***Exponent*** $= 5$

**Problem 3** *Identify the base and exponent of* $\left(\frac{1}{7}\right)^2$.

***Base*** $= \frac{1}{7}$ ***Exponent*** $= 2$

**Problem 4** *Write the expression* $11 \cdot 11$ *using exponential notation.*

***Multiple Choice:***

(a) $2 \cdot 11$

(b) $121$

(c) $11^2$ ✓

**Problem 5** *Write the expression* $9 \cdot 9 \cdot 9 \cdot 9$ *using exponential notation.*

***Multiple Choice:***

(a) $9 \cdot 4$

(b) $9^4$ ✓

(c) $9^3$

**Problem 6** *Write the expression* $2^5$ *as a product without the exponent.*

***Multiple Choice:***

(a) $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ ✓

(b) $2 \cdot 5$

(c) $2 \cdot 2 \cdot 2 \cdot 2$

**Problem 7** *Write the expression* $\left(\frac{1}{5}\right)^3$ *as a product without the exponent.*

***Multiple Choice:***

(a) $\left(\frac{1}{5}\right) \cdot 3$

(b) $\left(\frac{1}{5}\right) \cdot \left(\frac{1}{5}\right)$

(c) $\left(\frac{1}{5}\right) \cdot \left(\frac{1}{5}\right) \cdot \left(\frac{1}{5}\right)$ ✓

## Reading Numbers in Exponential Notation

Below are some powers of 3 and how they are read in exponential notation.

| | | | |
|---|---|---|---|
| $3$ | $=$ | $3^1$ | "Three or three to the first power." |
| $3 \cdot 3$ | $=$ | $3^2$ | "Three to the second power or three **squared**." |
| $3 \cdot 3 \cdot 3$ | $=$ | $3^3$ | "Three to the third power or three **cubed**." |
| $3 \cdot 3 \cdot 3 \cdot 3$ | $=$ | $3^4$ | "Three to the fourth power." |
| $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$ | $=$ | $3^5$ | "Three to the fifth power." |
| | $\vdots$ | | |

**Problem 8** *Write the expression below in words.*

$$100^3$$

***Multiple Choice:***

(a) *One hundred three*

(b) *Three hundred.*

(c) *One hundred times three.*

(d) *One hundred cubed.* ✓

**Problem 9** *Write the expression below in words.*

$$10^2 \cdot 3^6$$

***Multiple Choice:***

(a) *Ten times 2 times 3 times 6.*

(b) *Ten squared times three to the sixth power.* ✓

(c) *Thirty to the twelfth power.*

(d) *Ten to the sixth power times three squared.*

## Positive Powers of 10

A **positive power of 10** is a power of 10 in which the exponent is a **positive integer**: $\{1, 2, 3, 4, \ldots\}$. Below are the first three positive powers of 10. **Notice that the exponent in each power of 10 matches the number of zeros in the result**. In general, if $n$ is a positive integer, then the value of $10^n$ is the number 1 followed by $n$ zeros. For example, below you can see that $10^3$ equals the number 1 followed by 3 zeros.

$$\begin{aligned} 10^1 &= 10 \rightarrow 1 \text{ zero} \\ 10^2 &= 100 \rightarrow 2 \text{ zeros} \\ 10^3 &= 1,000 \rightarrow 3 \text{ zeros} \end{aligned}$$

**Problem 10** *Write the value of* $10^4$.

***Multiple Choice:***

(a) 40

(b) 10,000 ✓

(c) $1,000$

(d) $40,000$

**Problem 11** *Write the value of* $10^6$. ***Do not include any commas in your answer***.

$$10^6 = 1000000$$

*What number is this?*

***Multiple Choice:***

(a) *One hundred thousand.*

(b) *One million.* ✓

(c) *One billion.*

(d) *Ten thousand.*

**Problem 12** *A greedy CEO demanded a pay package worth \$1 trillion* $= 10^{12}$. *Write the value of* $10^{12}$.

***Multiple Choice:***

(a) $120$

(b) $10,000,000,000$

(c) $1,000,000,000$

(d) $1,000,000,000,000$ ✓

## Extra Insight: The Name Behind the Search

A **googol** is $10^{100}$, the number 1 followed by 100 zeros! In 1920, mathematician Edward Kasner was writing a book called *Mathematics and Imagination* and wanted to give a name to $10^{100}$. He asked his 9-year-old nephew, Milton Sirotta, for a suggestion and Milton came up with the word googol. Decades later, it inspired the name of one of the largest and well known tech companies: Google.

# Multiplying and Dividing by Positive Powers of 10

When we multiply or divide a number by a positive power of 10, there is a relationship between the movement of the decimal point and the number of zeros in the power of 10. **Once you notice this relationship, you won't need to perform multiplication or long division to find the answers to these types of problems**.

## Multiplying by Positive Powers of 10

Look for the relationship between the number of zeros in the power of 10 and the movement of the decimal point in the multiplication problems shown below.

$$\begin{aligned} 0.625(10) &= 6.25 && \text{The decimal point moved 1 place to the right.}\\ 0.625(100) &= 62.5 && \text{The decimal point moved 2 places to the right.}\\ 0.625(1000) &= 625 && \text{The decimal point moved 3 places to the right.} \end{aligned}$$

$$\begin{aligned} 5(10) &= 50 && \text{The decimal point moved 1 place to the right.}\\ 5(100) &= 500 && \text{The decimal point moved 2 places to the right.}\\ 5(1000) &= 5000 && \text{The decimal point moved 3 places to the right.} \end{aligned}$$

**Problem 13** *Write a general rule for multiplying a number by a positive power of 10.*

***Multiple Choice:***

(a) *To multiply a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the **left** that many places.*

(b) *To multiply a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the **right** that many places.* ✓

YouTube link: `https://www.youtube.com/watch?v=CTfY61uLv80`

**Problem 14** *Use the rule from the previous problem to multiply.*

2.8(10)= 28

$$3.0525(1000) = 3052.5$$

$$620(100) = 62000$$

## Dividing by Positive Powers of 10

Look for the relationship between the number of zeros in the power of 10 and the movement of the decimal point in the division problems shown below.

$\frac{43.2}{10} = 4.32$ The decimal point moved 1 place to the left.

$\frac{43.2}{100} = 0.432$ The decimal point moved 2 places to the left.

$\frac{43.2}{1000} = 0.0432$ The decimal point moved 3 places to the left.

$\frac{3}{10} = 0.3$ The decimal point moved 1 place to the left.

$\frac{3}{100} = 0.03$ The decimal point moved 2 places to the left.

$\frac{3}{1000} = 0.003$ The decimal point moved 3 places to the left.

**Problem 15** *Write a general rule for dividing a number by a positive power of 10.*

***Multiple Choice:***

(a) *To divide a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the* ***left*** *that many places.* ✓

(b) *To divide a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the* ***right*** *that many places.*

YouTube link: `https://www.youtube.com/watch?v=ne5Jyg6OK7w`

**Problem 16** *Use the rule from the previous problem to divide.*

$$\frac{63}{1000} = 0.063$$

$$\frac{35.5}{10} = 3.55$$

$$\frac{1}{100} = 0.01$$

[font=12pt]

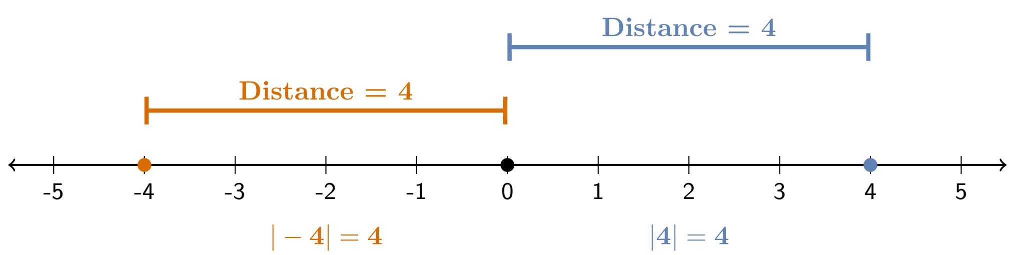
# 2 Absolute Value and Opposites- This sucks, nothing works!

## Absolute Value

If you're cutting a piece of wood to be 10 inches long, being 0.5 inches too long is just as problematic as being 0.5 inches too short. In both cases, the **size** of the error is 0.5 and this is what matters. The concept of **absolute value** can be used to represent the **magnitude** of a mistake, regardless of whether it was an overshot or undershot.

The **absolute value of a number** is its **distance from zero** on a number line. Since distance is always positive or 0, so is absolute value. Absolute value is represented by two vertical bars and can be interpreted as the **size**, **strength** or **magnitude** of a number.

The number line below shows the absolute values of $-4$ and 4. Since their absolute values are the same, the numbers $-4$ and 4 have the same strength or magnitude.




Alt text: A horizontal number line from negative 5 to 5 with solid dots plotted at negative 4, 0, and 4. Above the line, an orange bar spans from negative 4 to 0 and a blue bar spans from 0 to 4, both labeled "Distance = 4". Below the line, it states that the absolute value of negative 4 is 4 and the absolute value of 4 is 4.

**To find the absolute value of a number**, think of the number line as a path and each integer as a step. The absolute value of a number is how many steps you would have to take to travel from the number to zero. For example, you would have to take 2 steps to travel from $-2$ to 0, so the absolute value of $-2$ is 2.

$$|-2| = 2$$

**Problem 17** *Find the absolute value of each number.*

$$|-1| = 1$$

$$|7| = 7$$

$$|-55| = 55$$

**Problem 18** *You are building a custom bookshelf and need to cut boards that are 36 inches long. A board will be usable if its length is off by no more than 0.25 inch of the target. Suppose you cut the first board 35.8 inches long. The difference between the actual board length and the target length is the error.*

$$\begin{aligned} \textbf{\textit{Error}} &= 35.8 - 36 \\ &= -0.2 \ \textit{inches} \end{aligned}$$

*The negative sign indicates that you cut the board too short. However, it only matters if the board is useable or not, so we take the* ***absolute value*** *to find the* ***distance*** *from the target length. This distance is called the* ***absolute error****.*

$$\begin{aligned} \textbf{\textit{Absolute Error}} &= |35.8 - 36| \\ &= |-0.2| \\ &= 0.2 \ \textit{inches} \end{aligned}$$

*Is this board usable?* ***Hint****: A board is usable if the absolute error is less than or equal to 0.25 inches.*

***Multiple Choice:***

(a) *Yes, the absolute error is less than 0.25 inches.* ✓

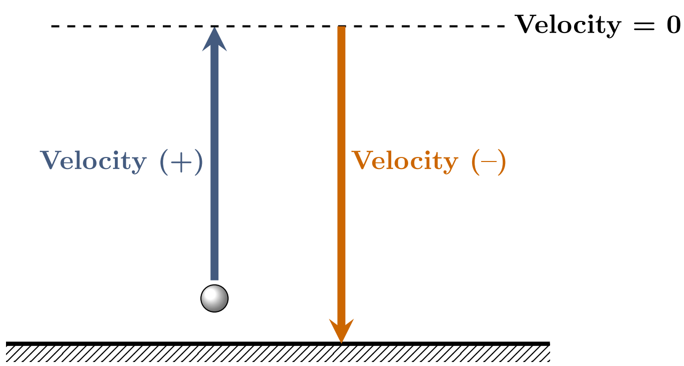
(b) *No, the absolute error is greater than 0.25 inches.*

## Extra Insight: The Relationship Between Speed and Velocity

In physics, the motion of an object is often described by its **velocity**, which indicates its speed *and* direction. When an object moves **upward** or to the **right**, its velocity is **positive** and when it moves **downward** or to the **left**, its velocity is **negative**. We use absolute value to turn velocity into speed.

Speed = —Velocity—
**Speed is the magnitude or size of velocity**.

When a ball is thrown straight up in the air, its velocity is positive as it moves upward, zero when it reaches its highest point and negative when it falls back to Earth. If, after 6 seconds, the velocity of the ball is $-15$ feet per second, we know that it is falling back to Earth, since its velocity is negative. Since $|-15| = 15$ the ball is traveling at a speed of 15 feet per second 6 seconds after it was thrown upward.




Alt text: A physics diagram illustrating vertical projectile motion relative to a ground line. A ball is above the ground on the left. A thick blue upward arrow represents the ball's ascending path and is labeled "Velocity" with a plus sign next to it. A dashed horizontal line marks the peak of the flight, labeled "Velocity = 0". A thick orange downward arrow on the right represents the descending path and is labeled "Velocity" with a minus sign next to it.

**Problem 19** *A model rocket was launched straight up in the air. After* 8 *seconds, the velocity of the rocket was recorded as* $-12$ *feet per second. In which direction was the rocket moving* 8 *seconds after it was launched?*

***Multiple Choice:***

(a) *Downward* ✓

(b) *Upward*

*What was the rocket's speed at that moment?*

*Speed* = 12 *feet per second*

## Opposites

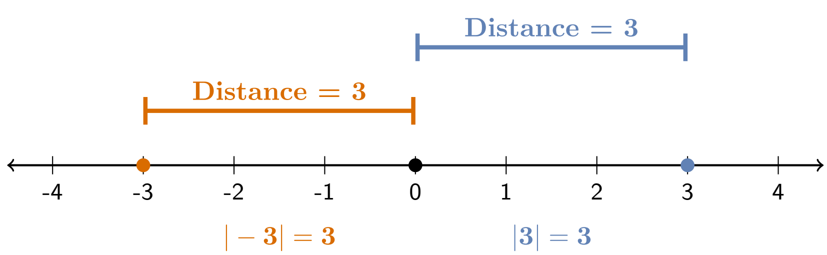
**Definition 1** (**Opposites**)**.** Two numbers that are the same distance from zero but on opposite sides of zero are called **opposites**.

The opposite of a number is found by changing its sign.

The phrase "**the opposite of**" is written mathematically as a dash ($-$).

**Example 1.** State two numbers that are opposites and explain why they are opposites.

**Explanation.** The numbers 3 and $-3$ are opposites. They lie the same distance from zero on the number line, but in **opposite** directions, and they have the same absolute values.




Alt text:
A horizontal number line from negative 4 to 4 with solid dots plotted at negative 3, 0, and 3. Above the line, an orange bar spans from negative 3 to 0 and a blue bar spans from 0 to 3, both labeled "Distance = 3". Below the line, it states that the absolute value of negative 3 is 3 and the absolute value of 3 is 3.

### Properties of Opposites

(a) Opposites have the same absolute value because they are equidistant from zero.

(b) One is positive and the other is negative.

(c) Adding a number and its opposite always results in zero: $a + (-a) = 0$.

(d) Zero is unique because it is its own opposite: $0 = -0$.

**Problem 20** *Complete the table.*

| *Number* | *Opposite* | *Absolute Value* |
|---|---|---|
| $-15$ | $15$ | $15$ |
| $8$ | $-8$ | $8$ |
| $0$ | $0$ | $0$ |
| $-2.5$ | $2.5$ | $2.5$ |

**Definition 2** (**The Three Meanings of a Dash** $-$)**.** In mathematics, the **dash** is used in three different ways depending on its role.

(a) **Negative**: When a dash is attached directly to a number, it describes its location on the number line and is **read** as **negative**.

(b) **Opposite**: A dash placed before parentheses, absolute value or a variable represents an action: changing the sign or multiplying by $-1$ and is **read** as **the opposite of**.

(c) **Subtraction**: A dash placed between two numbers represents the operation of subtraction and is **read** as **minus**.

**Remark 1.** The expression $-x$ is sometimes read as negative $x$, rather than the opposite of $x$. However, keep in mind that the expression can be positive, negative or even 0, depending on the value of $x$.

**Problem 21** *Identify the correct way to read each expression.*

(a) $18 - 5$ *is read as (Negative 5/ Eighteen minus five ✓/ The opposite of 5).*

(b) $-20$ *is read as (Negative 20 ✓/ The opposite of negative 20/ Minus 20).*

(c) $-|-7|$ *is read as (The opposite of the absolute value of negative 7 ✓/ The opposite of the absolute value of 7. / The absolute value of negative 7).*

(d) $-k$ *is read as (Dash k/ Minus k/ The opposite of k ✓)*

**Definition 3 (The Double Negative Property).** For any number $a$,

$$-(-a) = a$$

The expression $-(-a)$ is read as "the opposite of negative $a$."

**Problem 22** *Use the Double Negative Property to simplify the following expressions.*

$$-(-14) = 14$$

$$-(-100) = 100$$

$$-(-0) = 0$$

**Problem 23** *How is the expression $-(-22)$ read?*

***Multiple Choice:***

(a) *Negative negative 22*

(b) *Minus negative 22*

(c) *The opposite of negative 22* ✓

(d) *The opposite of 22*

**Problem 24** *If $x = -10$, what is the value of $-(-x)$?*

***Multiple Choice:***

(a) 10

(b) $-10$ ✓

(c) 0

***Feedback(`attempt`):*** *Remember: $-(-x) = x$, therefore since $x$ started as $-10$, the final result is also $-10$.*

**Warning 1.** A common mistake is to treat absolute value like parentheses. Keep in mind that **parentheses** are used for **grouping**, whereas **absolute value** is an **operation**. Make certain you understand the difference between the following cases.

**Double Negative with Parentheses**
Since we have parentheses, we simply find the opposite of negative 7, which is 7.

$$-(-7) = 7$$

**Double Negative with Absolute Value**
To find the opposite of the absolute value of negative 7, start with the absolute value: $|-7| = 7$, then find the opposite of that result. The opposite of 7 is $-7$ therefore,

$$-|-7| = -7.$$

## Absolute Value vs. Opposite

While absolute value always results in a nonnegative (zero or positive) number, taking the opposite of a number changes its sign. Therefore, an opposite can be positive, negative or 0. Note that for the number 0, both its absolute value and its opposite are 0. Later, you will see how these two concepts are used together to add integers.

Below, the concept of an opposite has been used to write a rule for finding absolute value. Notice that this rule doesn't rely on distance.

**Procedure 1 (Finding the Absolute Value of a Number). Positive Numbers and 0:**
The absolute value of a positive number or 0 is the number itself.

**Negative Numbers:**
The absolute value of a negative number is the **opposite** of that number.

**Problem 25** ***True or False*** *If a number is negative, its opposite must be positive.*

***Multiple Choice:***

(a) *True* ✓

(b) *False*

***Feedback(*`correct`*):*** *Exactly! The opposite flips the sign, so a negative becomes positive.*

**Problem 26** ***True or False*** *Two different numbers can have the same absolute value.*

***Multiple Choice:***

(a) *True* ✓

(b) *False*

***Feedback(*`correct`*):*** *Yep! The numbers* $2$ *and* $-2$ *have the same absolute values. In fact, all opposites have the same absolute values.*

**Problem 27** *Which has a greater value: the opposite of* $-10$ *or the absolute value of* $-10$*?*

***Multiple Choice:***

(a) *Neither, they are the same.* ✓

(b) *The opposite of* $-10$*.*

(c) *The absolute value of* $-10$*.*

***Feedback(*`attempt`*):*** *The opposite of* $-10$ *is* $10$ *and* $|-10| = 10$*, so they are the same.*

**Problem 28** *Two numbers, $a$ and $b$ are opposites. If the distance between $a$ and $b$ on a number line is $10$ units, what are the two numbers?*

***Multiple Choice:***

(a) $-10$ *and* $10$

(b) $-5$ *and* $5$ ✓

(c) $-6$ *and* $4$

(d) $-7$ *and* $3$

***Hint:*** *Draw a number line and with $a$ and $b$ ten units apart.*

***Feedback(*`correct`*):*** *Good job! You made it through this problem and this section.*

[font=12pt]

# 3 Adding Integers: Introduction

## The Set of Integers

In this lesson, you will learn how to add numbers that belong to the set of integers. The set of **integers** consists of the whole numbers along with the negative counting numbers.

Counting Numbers: $\{1, 2, 3, \dots\}$

Whole Numbers: $\{0, 1, 2, 3, \dots\}$

Integers: $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

## Extra Insight: False Numbers

Although negative numbers were understood in China more than 2000 years ago and later in India, they were not immediately accepted in Europe. For many centuries, European mathematicians were uncomfortable with the idea of numbers less than zero and sometimes referred to them as "false" or "absurd" numbers. However, during the 16th and 17th centuries, as algebra and calculus developed, the importance of negative numbers became undeniable. Today, negative numbers are an essential part of mathematics and are commonly used to represent quantities such as losses, debts, temperatures below zero and elevations below sea level.

## Adding Integers

To understand how to add integers, we will begin with the Chinese Red-Black Rule and then a number line to visualize addition. Then we will use the concepts of assets and debts to understand addition in a real-world context. These approaches will lead us to two rules: one for adding numbers with the same sign and another for adding numbers with different signs.

## The Red-Black Rule

Negative numbers were first used systematically in ancient China. In *The Nine Chapters on the Mathematical Art* (written between about 200 BC and 100 AD), Chinese mathematicians used counting rods to represent numbers. Red rods were used for positive numbers and black rods for negative numbers. For example, the number 3 was represented by three red rods and the number $-2$ was represented by two black rods. These rods were used as a calculator of sorts for financial transactions involving profits and losses and to solve simple equations. The Red-Black Rule is described below.

(a) A red rod represents 1 and a black rod $-1$.

(b) To find the total of profits (red) and losses (black), begin by placing rods on a board to represent each number.

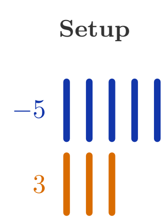
(c) Since a profit and an equal loss result in 0, gather all pairs of red and black rods and remove them.

(d) The total is what remains.

Since red and black can be difficult for some people to distinguish, we will use **orange rods** to represent **positive numbers (+)** and **blue rods** to represent **negative numbers (−)**. We will now refer to this rule as the Orange-Blue Rule.

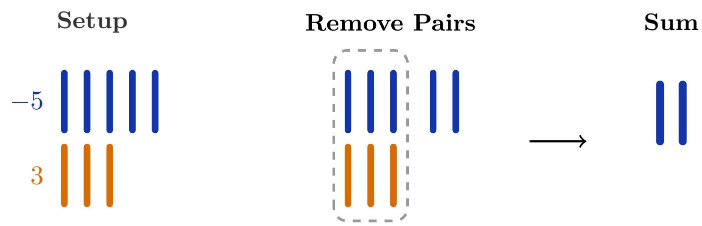
**Example 2.** Find the sum $-5 + 3$ using the Orange-Blue Rule.

**Explanation.** We begin with **five blue rods** to represent the number $-5$ and **three orange rods** to represent the number 3.




Alt text: The "Setup" section showing five blue vertical bars representing negative 5 and below them are three orange vertical bars representing 3.

Next, we remove all pairs of orange and blue rods. The sum is what remains.



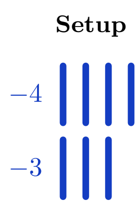

Alt text: A diagram modeling the addition of a negative number and a positve number using vertical color-coded rods. In the "Setup" section, on the left, five blue bars represent $-5$ and below are three orange bars representing 3. In the "Remove Pairs" section, in the middle, the three orange bars are paired vertically with three of the blue bars inside a dashed box to show they cancel each other out, leaving two blue bars outside the box. A black arrow points to "Sum" section which shows the two remaining blue bars.

Count the number of bars that remain to find the sum. Remember to use the correct sign.

$-5 + 3 = -2$

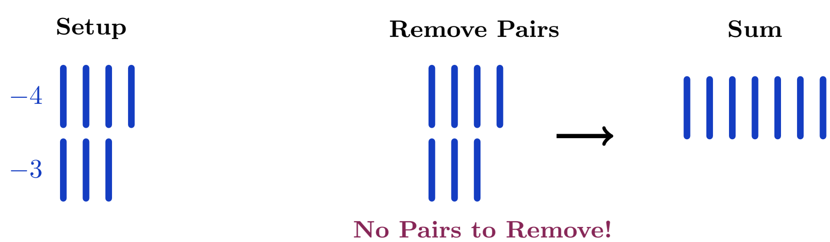
**Example 3.** Find the sum $-4 + (-3)$ using the Orange-Blue Rule.

**Explanation.** We begin with **four blue rods** to represent the number $-4$ and **three blue rods** to represent the number $-3$.




Alt text: The "Setup" section showing four blue vertical bars representing negative 4 and below them are three blue vertical bars representing negative 3.

Since we started with only blue rods, there are no orange-blue pairs to remove.




Alt text: A diagram modeling the addition of two negative numbers using vertical color-coded rods. In the "Setup" section, on the left, there are four blue bars representing negative 4 and below are three blue bars representing negative 3. The "Remove Pairs" section, in the middle, is the same as the "Setup" section. A caption below the rods reads "No Pairs to Remove!" because all rods are the same color. A black arrow points to the "Sum" section, which shows all seven blue rods together.

Count the number of bars that remain to find the sum. Remember to use the correct sign.

$-4 + (-3) = -7$

**Problem 29** *Use the Blue-Orange Rule to find the sum $-2+6$. The problem has been started for you below.*


*Alt text: A diagram modeling the addition of a negative number and a positive number using vertical color-coded rods. In the "Setup" section, on the left, there are two blue vertical bars representing negative 2 and six orange rods representing 6. The "Remove Pairs" section, on the right, is the same as the "Setup" section.*

*Count the number of bars that remain to find the sum. Remember to use the correct sign.*

$-2 + 6 = 4$

The use of colors in the Orange-Blue Rule is not necessary. Below, the first example has been redone using a plus sign (+) for positive 1 and a minus sign (−) for −1. You can also use things like rocks and sticks or pennies and dimes—the sky's the limit! Try modifying the rule, using your own way to represent positive and negative numbers. Be creative!

**Example 4.** Find the sum $-5 + 3$ using a plus sign (+) for 1 and a minus sign (−) for −1.

**Explanation.** We begin with **five minus signs** (−) to represent the number −5 and **three plus signs** (+) to represent the number 3.

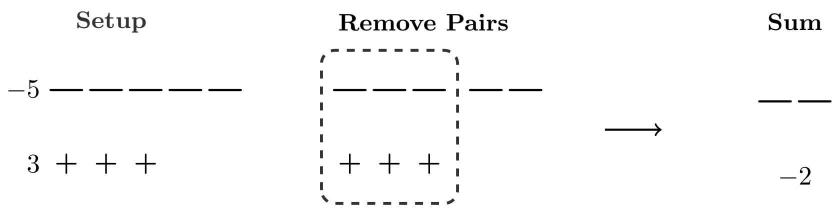
**Setup**

−5 — — — — —

3 + + +

Alt text: The 'Setup' showing five minus signs to represent −5 and below three plus signs represent 3.

Next, we remove all plus-minus pairs. The sum is what remains.




Alt text: A diagram modeling the addition of a negative number and a positive number using plus and minus signs. In the “Setup” section, on the left, five minus signs represent negative 5 and below them three plus signs represent 3. In the “Remove Pairs” section, in the middle, three minus signs and three plus signs are grouped inside a dashed box to show that they cancel. Two minus signs remain outside the box. A black arrow points to the “Sum” section which shows two minus signs labeled negative 2.

$-5 + 3 = -2$

Although this modified rule isn’t overly creative, it makes the answer quite clear.

[font=12pt]

## 3.1 Adding Integers: Numberline

### Adding Integers on a Number Line

Another way to visualize addition is to add on a number line.

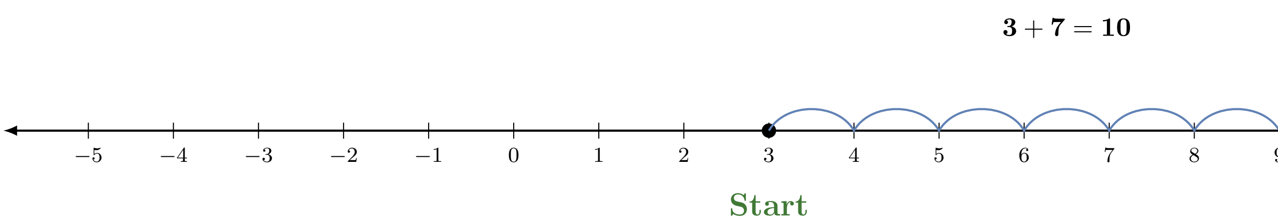
**Procedure 2** (**Steps for Adding Integers on a Numberline**). (a) Locate the first number on the number line.

(b) If the second number is **positive**, move to the **right** that many units.

(c) If the second number is **negative**, take the absolute value of the number and move to the **left** that many units.

**Example 5.** Use a numberline to find the sum $3 + 7$.

**Explanation.** Start at 3, then move 7 units to the **right**. Movement is to the right, since a **positive** number is added to 3.

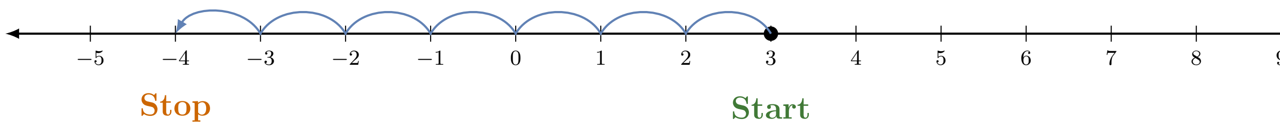



Alt text: Horizontal number line from negative 5 to 10, with tick marks and labels for each integer. There is a solid black dot at 3. The word "Start" is written below the number 3 in green. A series of connected, blue arches move rightward from 3, landing on each integer until stopping with a right-pointing arrow at 10. The word "Stop" is written below the number 10 in orange. The equation "3 plus 7 equals 10" is displayed centered above the arches.

**Example 6.** Use a numberline to find the sum $3 + (-7)$.

**Explanation.** Start at 3 then move $|-7| = 7$ units to the **left**. Movement is to the left, since a **negative** number is added to 3.

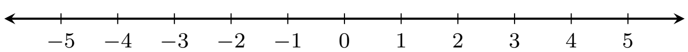
$$\mathbf{3 + (-7) = -4}$$




Alt text: Horizontal number line from negative 5 to 10, with tick marks and labels for each integer. There is a solid black dot at 3. The word “Start” is written below the number 3 in green. A series of connected, blue arches move leftward from 3, landing on each integer until stopping with a left-pointing arrow at negative 4. The word “Stop” is written below the number negative 4 in orange. The equation “3 plus negative 7 equals negative 4” is displayed centered above the arches.

**Problem 30** *Add using the number line shown below.*

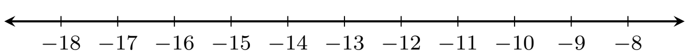
$$-2 + 3 = 1$$




*Alt text: Horizontal number line from $-5$ to 5, with tick marks and labels for each integer.*

**Problem 31** *Add using the number line shown below.*

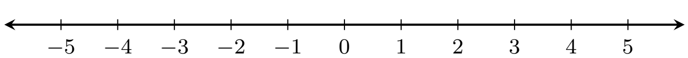
$$-12 + (-3) = -15$$




*Alt text: Horizontal number line from negative 18 to negative 8, with tick marks and labels for each integer spanning the full width.*

**Problem 32** *Add using the number line shown below.*

$$-1 + 4 = 3$$




*Alt text: Horizontal number line from negative 5 to 5, with tick marks and labels for each integer.*

**Problem 33** *Does the answer to the previous problem change if you switch the order of the numbers and find the value of* $4 + (-1)$?

***Multiple Choice:***

(a) *No* ✓

(b) *Yes*

***Feedback(*`correct`*):*** *That's right, the answer does NOT change! This is due to a property of addition called the Commutative Property. The* ***Commutative Property of Addition*** *states that numbers can be added in any order and the result will be the same. Therefore,* $-1 + 4 = 4 + (-1)$.

[font=12pt]
## 3.2 Adding Integers: Financial Reasoning

### Adding Integers Using Financial Reasoning

Number lines and the Orange-Blue Rule are great for visualizing addition and finding answers to simple problems, but they aren't always practical. For example, we would need a very large number line to find the value of $-185 + 99$. Instead, we can add integers by thinking in terms of money.

- **Assets (+)** include money that you **have**, a gift you receive or a deposit.
- **Debts (−)** include money that you **owe**, a bill you pay or a withdrawal.

When we add integers that represent assets and debts, we are finding the **net balance**.

$$\textbf{Net Balance} = \text{Assets} + \text{Debts}$$

**Remark 2.** A statement such as "a debt of \$50" is represented mathematically as $-50$. The negative sign is omitted in the phrase because the word *debt* already implies a negative value.

**Example 7.** Combining Assets

If a child has \$100 (+100) in her piggy bank and receives \$30 (+30) for doing her chores, her net balance is **\$130**.

$$100 + 30 = 130$$

**Example 8.** Combining Debts

If you have a credit card debt of \$500 (−500) and make another credit card purchase of \$100 (−100), your net balance is a **debt of \$600 (**−600**)**.

$$-500 + (-100) = -600$$

**Example 9.** Partial Cancellation

If you have \$20 (+20) and owe a friend \$25 (−25). Your \$20 isn't enough to cover the entire debt, it only cancels **part** of it. After you pay your friend \$20, you will still owe \$5, so your net balance is a **debt of \$5 (**−5**)**.

$$20 + (-25) = -5$$

**Example 10.** Full Cancellation

If a taxpayer owes the IRS \$300 (−300) and has \$800 (+800) in the bank, they can pay the entire debt and keep the change. Their net balance is **\$500 (**+500**)**.

$$-300 + 800 = 500$$

### Key Observations

If you take a close look at the examples above you will notice the following relationships between the signs of the numbers that were added and the operation we actually performed.

> **Same Signs**: When we add two assets or two debts, we **add** their absolute values. The sign of the answer is the same as the sign of the numbers.
>
> **Different Signs**: When we add an asset and a debt, all or part of the debt will be removed or cancelled, so even though we are technically *adding* integers, the arithmetic performed is **subtraction**. The sign of the answer depends on which is greater: the asset or the debt.

**Problem 34** *Your checking account is overdrawn by \$50 and the bank charges you a \$15 overdraft fee. Write the addition problem to find your net balance. Then calculate your net balance. If the net balance is a debt, use a negative sign in your answer.*

***Multiple Choice:***

(a) $50 + (-15)$

(b) $-50 + 15$

(c) $50 + 15$

(d) $-50 + (-15)$ ✓

***Net Balance*** $= -65$

**Problem 35** *A soccer team spent \$25 on supplies for a car wash. By the end of the day, they earned \$175. Write the addition problem to find the team's net balance. Then calculate their net balance. If the net balance is a debt, use a negative sign in your answer.*

***Multiple Choice:***

(a) $25 + 175$

(b) $-25 + 175$ ✓

(c) $-25 + (-175)$

(d) $25 + (-175)$

***Net Balance*** $= 150$

**Problem 36** *It's your birthday! You receive \$300 as a gift, but you owe your roommate \$500 for rent. Write the addition problem to find your net balance after you use your gift money toward the rent. Then calculate your net balance. If the net balance is a debt, use a negative sign in your answer.*

***Multiple Choice:***

(a) $-300 + (-500)$

(b) $300 + 500$

(c) $300 + (-500)$ ✓

(d) $-300 + 500$

***Net Balance*** $= -200$

**Problem 37** *Think in terms of assets and debts to find the following sums.*

$$80 + (-2) = 78$$

$$-9 + (-1) = -10$$

$$-30 + 27 = -3$$

[font=12pt]

## 3.3 Adding Integers: Different Signs

### Adding Integers with Different Signs

When adding integers with different signs, we can think of the process as a **battle between positive and negative numbers**, where the winner is the sign of the answer.

**Battle 1:** If you have \$30 and spend \$20 to see a movie, your net balance is \$10.

$$30 + (-20) = 10$$

The answer is positive, so the positive team win!

**Battle 2:** If you have \$30 and a credit card debt of \$100, your net balance is a debt of \$70.

$$30 + (-100) = -70$$

The answer is negative, so the negative team win!

**Why did the positive team win the first battle and lose the second battle?**

The positive team won the first battle because the asset was greater than the debt, but lost the second because the debt was greater than the asset. In any number battle, the winner is always the team with the **greatest strength**, regardless of its sign. Mathematically, we find the **absolute value** of a number to determine its **magnitude**, **strength** or **size**.

**Battle 1:** $30 + (-20) = 10$

The strength of the \$30 asset is $|30| = 30$.

The strength of the \$20 debt is $|-20| = 20$.

Since $|30|$ is larger than $|-20|$, the **positive** team was stronger and **won** the battle.

**Battle 2:** $30 + (-100) = -70$

The strength of the \$30 asset is $|30| = 30$.

The strength of the \$100 debt is $|-100| = 100$.

Since $|-100|$ is larger than $|30|$, the **negative** team was stronger and **won** the battle.

We are now ready for a formal mathematical rule that can be applied to any problem, not just those involving money. As noted earlier, adding numbers with different signs is a bit of a paradox: to find the sum, we actually have to subtract. From the number battles, we know that the sign of the answer depends on which number has the greatest strength. To turn these observations into a formal rule, we use absolute value.

**Procedure 3** (Steps for Adding Integers with Different Signs)**.** (a) Find the **absolute value** or **strength** of each number.

(b) Subtract the smaller absolute value from the larger absolute value.

(c) Write the answer with the sign of the number with the **larger absolute value**.

**Example 11.** Use the steps to find the sum $-50 + 30$.

**Explanation. Step 1: Find the Absolute Values (The Strengths)**
Find the strength of each number.

$$|-50| = 50 \quad \text{and} \quad |30| = 30$$

**Step 2: Subtract the Absolute Values (The Battle)**
Since we are adding numbers with different signs, the numbers "battle" by finding the difference between their strengths.

$$50 - 30 = 20$$

**Step 3: Declare the Winner and Write the Answer (The Sign)**
Compare the absolute values from the first step. Since $|-50|$ is greater than $|30|$, the **negative team** is stronger and **wins** the battle!

$$-50 + 30 = -20$$

**Problem 38** *Use the steps to find the sum* $15 + (-40)$.

**Step 1:** ***Find the Absolute Values (The Strengths)***

$$|15| = 15 \quad \textit{and} \quad |-40| = 40$$

**Step 2:** ***Subtract the Absolute Values (The Battle)***

$$40 - 15 = 25$$

**Step 3:** ***Declare the Winner and Write the Answer (The Sign)***
*Compare the absolute values: is $|15|$ or $|-40|$ larger?*

***Multiple Choice:***

(a) *The positive team wins, since $|15|$ is larger.*

(b) *The negative team wins, since $|-40|$ is larger.* ✓

$15 + (-40) = -25$

**Problem 39** *Find the following sums.*

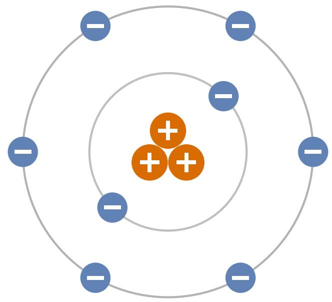
$$-12 + 4 = -8$$

$$12 + (-4) = 8$$

$$-20 + 32 = 12$$

$$100 + (-600) = -500$$

**Problem 40** *Atoms are the tiny building blocks of all matter. An atom consists of protons, electrons and neutrons. Each proton has a positive charge (+1), each electron has a negative charge (−1) and neutrons have no charge (0). Chemists find the charge of an atom by adding the charges of its electrons and protons. Write the addition problem to find the charge of an atom that has eight electrons and three protons. Then find the charge of the atom.*



*Alt text: A model of the atom. The nucleus has 3 large orange protons with white plus signs. Two gray orbits surround it: an inner orbit with 2 blue electrons and a slightly larger outer orbit with 6 blue electrons. All electrons have white minus signs.*

***Multiple Choice:***

(a) $-8+3$ ✓

(b) $-8+(-3)$

(c) $8+(-3)$

(d) $8+3$

***Charge*** $= -5$

[font=12pt]

## 3.4 Adding Integers: Same Signs

### Adding Integers with the Same Sign

When adding numbers with the same sign, there is no "battle." Instead, the numbers **join forces** to create a larger asset or a deeper debt.

**Example 12.** **Joining Assets**

If you have \$400 in your savings account (+400) and you win \$150 (+150), you have a net balance of \$550.

$$400 + 150 = 550$$

**Example 13.** **Joining Debts**

If you owe your brother \$10 ($-10$) and then borrow another \$25 ($-25$), your net balance is a debt of \$35. So, you are falling deeper into debt.

$$-10 + (-25) = -35$$

We can now write a formal mathematical rule for adding integers with the same sign. Once again, we will use absolute value.

**Procedure 4** (**Steps for Adding Integers with the Same Signs**). (a) Find the **absolute value** or **strength** of each number.

(b) **Add** the absolute values.

(c) Write the answer with the common sign.

**Example 14.** Use the three-step process to find the sum $-50 + (-30)$.

**Explanation.** **Step 1** **Find the Absolute Values (Strengths)**:
Fnd the strength of each number.

$$|-50| = 50 \quad \text{and} \quad |30| = 30$$

**Step 2** **Add the Absolute Values (Join Forces)**:
Since the signs are the same, we add their absolute values.

$$50 + 30 = 80$$

**Step 3** **Determine the Sign and Write the Answer**:
Since both numbers are negative, the answer is negative.

$$-50 + (-30) = -80$$

**Problem 41** *Apply the steps to find the sum $-15 + (-40)$.*

**Step 1** ***Find the Absolute Values (Strengths)***:

$$|-15| = 15 \quad \text{and} \quad |-40| = 40$$

**Step 2** ***Add the Absolute Values (Join Forces)***:

$$15 + 40 = 55$$

**Step 3** ***Determine the Sign and Write the Answer***:

$$-15 + (-40) = -55$$

**Problem 42** *A research submarine is exploring the ocean floor. It starts at a depth of 5 meters below sea level. The pilot then navigates the submarine down another 8 meters. Find the submarine's final depth relative to sea level.*

$$-5 + (-13) = -18 \text{ meters}$$

**Problem 43** *During a football game, the home team's running back gained 6 yards on the first play of the drive. On the very next play, he charged forward an additional 7 yards. Find the total number of yards gained by the running back over these two plays.*

$$6 + 7 = 13 \text{ yards}$$

**Problem 44** *On a cold winter night in Cleveland, the starting temperature at midnight was* $-1^\circ F$*. Over the next three hours, another cold front moved through the area and the temperature dropped by* $9^\circ F$*. Find the temperature at 3am.*

$$-1 + (-9) = -10\ ^\circ F$$

[font=12pt]

## 3.5 Adding Integers: Commutative Property

### The Commutative Property of Addition

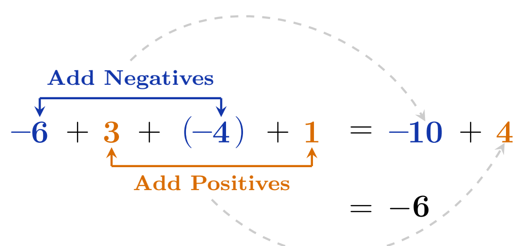
The operation of addition is **commutative**, meaning numbers can be added in any order without changing the result. For example,

$$4 + 3 = 3 + 4.$$

When adding multiple numbers with different signs, we can use this property to work more efficiently. Instead of working from left to right, simply combine the positive numbers and the negative numbers separately, then add those results.

**Example 15.** Use the commutative property to find the sum $-6+3+(-4)+1$.

**Explanation.** Add the negative numbers $-6$ and $-4$, then add the positive numbers 3 and 1. Then add the resulting integers to find the answer.




Alt text: Diagram showing how to use commutative property to add negative 6 plus 3 plus negative 4 plus 1. A blue line connects negative 6 and negative 4 with the label "Add Negatives". An orange line connects 3 and 1 with the label "Add Positives". Dashed arrows show these groups simplifying on the right side to negative 10 plus 4. Below this step, the final line shows an equals sign and the simplified result, negative 6.

**Problem 45** *Use the commutative property to find each sum.*

$$-5 + 2 + (-4) + 8 = 1$$

$$3 + (-1) + 6 + (-15) + 1 = -6$$

$$7 + (-12) + 3 = -2$$

**Great News!**

The rules for adding integers apply to all **real numbers**. The integers, along with the fractions and decimals between them, make up the set of real numbers—more on that later. Did you know that there are infinitely many numbers between each integer?

Once you have mastered these addition rules, you will be ready for subtraction. As you will see in the next lesson, **if you can add integers, you can subtract them too!**

[font=12pt]

# 4 Subtracting Integers: Introduction

## Introduction

**Elevation** is the height of a point on earth **above** (+) or **below** (−) sea level. In geology, the **elevation span or topographic relief** of a region is the difference between its highest and lowest elevations. Regions with large spans are typically rugged, with high mountain ranges and deep valleys, whereas regions with small spans are generally flatter and more uniform.

$$\textbf{Elevation Span} = \text{Highest Elevation} - \text{Lowest Elevation}$$

By calculating the difference between the highest and lowest elevations, geologists can determine if a region is being built up by internal forces (large span) or worn away by the elements (small span).

When both points of interest are above sea level, finding the span is easy. In Ohio, for example, the highest point (Campbell Hill) is 1550 feet **above** sea level, and the lowest point (Ohio River) is 455 feet **above** sea level, so the elevation span of Ohio is

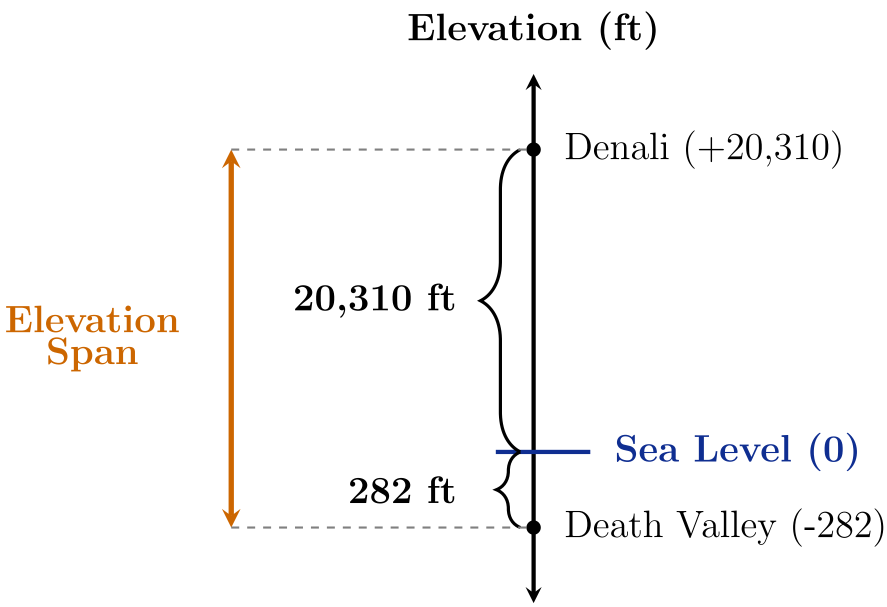
$$1550 - 455 = 1095 \text{ feet.}$$

When one elevation is below sea level, finding the span isn't as straightforward. For instance, in the United States, the highest peak is 20,310 feet **above** sea level in Denali, Alaska and the lowest basin is 282 feet **below** sea level in Death Valley, California. So, the elevation span of the U.S. is given by the difference

$$20,310 - (-282).$$

However, from the diagram below, you can see that the span of the U.S. can be found by *adding*.

$$20,310 + 282 = 20,592 \text{ feet}$$




Alt text: Vertical number line showing the elevation span for the United States. Points are plotted at 20,310 for Denali and negative 282 for Death Valley, with sea level marked at zero. Braces indicate a distance of 20,310 units above zero and 282 units below zero. The total elevation span is shown to the left as the combined vertical distance from negative 282 to 20,310.

Although it seems paradoxical, this illustrates a key concept: **subtracting a number is the same as adding its opposite**.

$$20,310 - (-282) = 20,310 + 282$$

**Problem 46** *China has the largest elevation span in the world. Its highest elevation is 29,032 feet **above** sea level on Mount Everest and its lowest elevation is 505 feet **below** sea level at Ayding Lake. Write the subtraction problem to find China's elevation span.*

***Multiple Choice:***

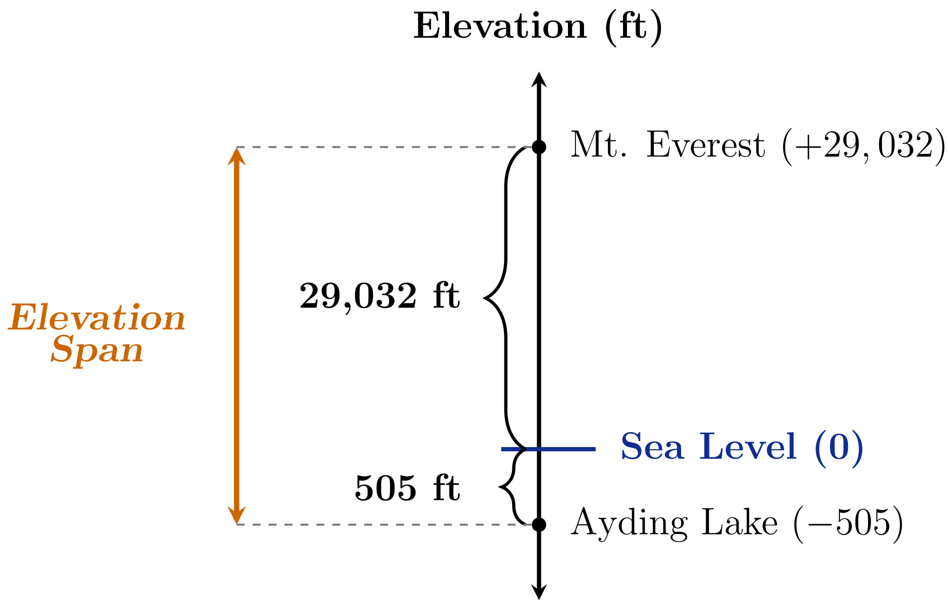
(a) $29,032 - 505$

(b) $-29,032 - (-505)$

(c) $29,032 - (-505)$ ✓

(d) $-29,032 - 505$

*Use the diagram below to write the subtraction problem as an equivalent addition problem. Then find China's elevation span.*




*Alt text: Vertical number line showing the elevation span for China. Points are plotted at 29,032 (Mount Everest) and negative 505 (Ayding Lake). Braces indicate a distance of 29,032 units above zero and 505 units below zero. The total elevation span is shown as the combined vertical distance from negativve 505 to 29,032.*

***Multiple Choice:***

(a) $-29,032 + (-505)$

(b) $29,032 + (-505)$

(c) $-29,032 + 505$

(d) $29,032 + 505$ ✓

***Do not include commas in your answer.***

***China's Elevation Span*** = 29537 *feet*

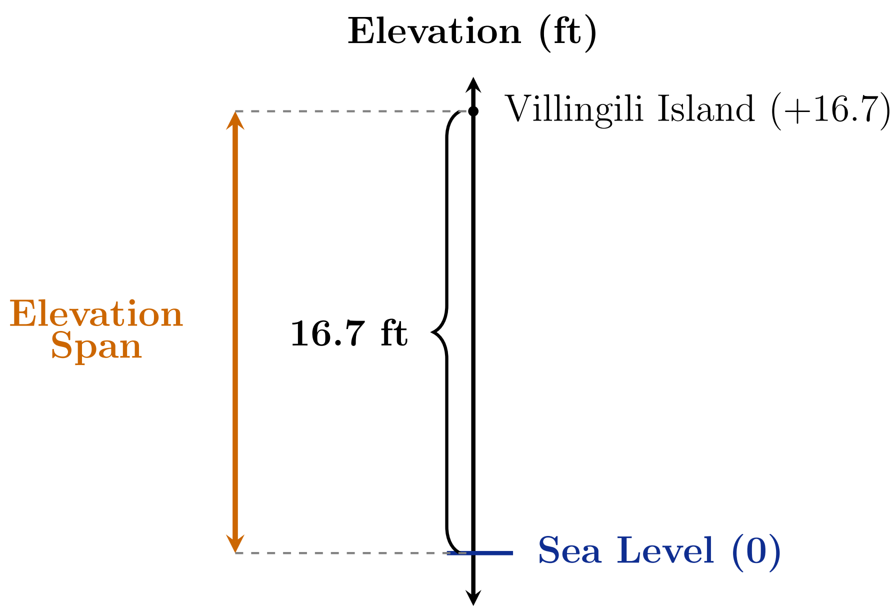
## Extra Insight: The Drowning of the Maldives

An **archipelagic nation** is a country made up of a group of islands and the surrounding water. The Maldives is an archipelagic nation of over 1,000 islands located in the Indian Ocean. It is the lowest-lying country on Earth, with an average ground level elevation of less than 5 feet above sea level. It also has the world's smallest elevation span: just 16.7 feet above sea level. The islands are very flat!

**Highest Elevation**: 16.7 feet **above** sea level (Villingili Island)

**Lowest Elevation**: 0 feet, **at** sea level (Indian Ocean)

Scientists predict that most of the Maldives will be uninhabitable or completely submerged by the year 2100 if sea levels continue to rise at current rates.




Alt text: Vertical number line showing the elevation span for the Maldives. A point is plotted at 16.7 for Villingili Island, and sea level is marked at zero. A brace indicates a distance of 16.7 feet above zero. The total elevation span is shown to the left as the vertical distance from zero to 16.7.

[font=12pt]

## 4.1 Subtracting Integers: Rule

### Rule for Subtracting Integers

Our work with elevation spans demonstrated that **subtracting a number is equivalent to adding its opposite**. This rule is stated formally below.

**Procedure 5** (**Subtracting Integers**)**.** If $a$ and $b$ are integers, then $a - b = a + (-b)$.

**Example 16.** Subtracting $-12$ is equivalent to adding positive 12.
$6 - (-12) = 6 + 12$

Subtracting positive 12 is equivalent to adding $-12$.
$6 - 12 = 6 + (-12)$

**Remark 3.** Once the subtraction problem has been rewritten as addition, use the rules for adding integers to find the answer.

**Remark 4.** When you subtract a positive number from a *larger* positive number, there is no reason to rewrite the subtraction problem as addition of the opposite. For example, you don't need to rewrite $7 - 2$ as $7 + (-2)$ to figure out that the answer is 5. So, **only rewrite subtraction as addition if you need to**.

**Problem 47** *Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.*

$$-6 - (-3)$$

***Multiple Choice:***

(a) $-6 - (-3) = 6 + 3$

(b) $-6 - (-3) = -6 + (-3)$

(c) $-6 - (-3) = -6 + 3$ ✓

(d) $-6 - (-3) = 6 + (-3)$

***Answer*** $= -3$

INSERT VIDEO!!!

**Problem 48** *Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.*

$$-2 - 10$$

***Multiple Choice:***

(a) $-2 - 10 = 2 + (-10)$

(b) $-2 - 10 = 2 + 10$

(c) $-2 - 10 = -2 + (-10)$ ✓

(d) $-2 - 10 = -2 + 10$

***Answer*** $= -12$

**Problem 49** *Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.*

$$5 - 8$$

***Multiple Choice:***

(a) $5 - 8 = -5 + 8$

(b) $5 - 8 = 5 + (-8)$ ✓

(c) $5 - 8 = -5 + (-8)$

(d) $5 - 8 = 5 + 8$

***Answer*** $= -3$

**Problem 50** *Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.*

$$-16 - (-1)$$

***Multiple Choice:***

(a) $-16-(-1)=-16+1$ ✓

(b) $-16-(-1)=-16+(-1)$

(c) $-16-(-1)=16+1$

(d) $-16-(-1)=16+(-1)$

***Answer*** $= -15$

**Problem 51** *Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.*

$$30-(-10)$$

***Multiple Choice:***

(a) $30-(-10)=30+(-10)$

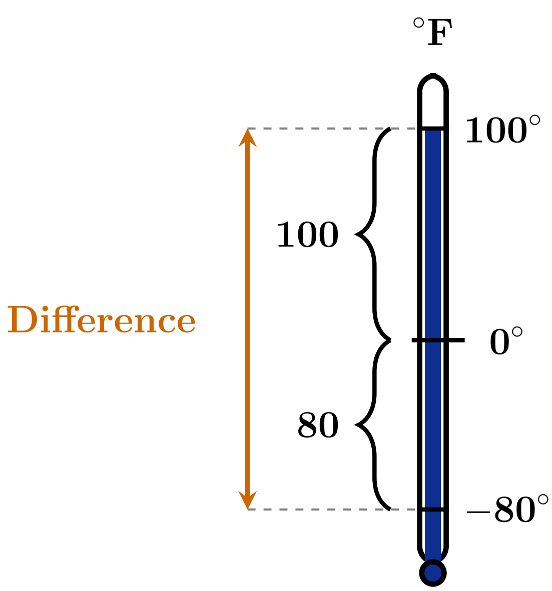
(b) $30-(-10)=30+10$ ✓

(c) $30-(-10)=-30+10$

(d) $30-(-10)=-30+(-10)$

***Answer*** $= 40$

**Problem 52** *How many degrees warmer is $100^\circ F$ than $-80^\circ F$?*



*Alt text: A vertical thermometer diagram labeled in degrees Fahrenheit, illustrating the distance between negative 80 degrees and positive 100 degrees. The blue liquid inside the thermometer reaches up to the 100-degree mark. A solid black line marks the 0-degree point in the middle. On the left side, visual brackets measure the distances from negative 80 to 0 (labeled 80) and from 0 to 100 (labeled 100). Further to the left, a double-headed orange arrow spans the entire height from negative 80 to 100, labeled "Difference" with dashed gray guide lines connecting it to the temperature marks.*

*Write the subtraction problem to find the answer.*

***Multiple Choice:***

(a) $100 - 80$

(b) $100 - (-80)$ ✓

(c) $-100 - 80$

(d) $-100 - (-80)$

*Use the figure to rewrite the subtraction problem as an equivalent addition problem. Then find the answer.*

***Multiple Choice:***

(a) $100 + (-80)$

(b) $-100 + 80$

(c) $-100 + (-80)$

(d) $100 + 80$ ✓

***Answer*** = $180\ ^\circ F$

### Subtracting a Negative Number

You may have noticed that **subtracting a negative** number is equivalent to **adding a positive** number. This becomes intuitive when we think in terms of money. Imagine your checking account balance is \$900 and the bank realizes that they accidently charged you a \$30 ($-30$) overdraft fee last month. Once the bank removes (subtracts) the overdraft fee, your net balance increases.

$$\begin{aligned} 900 - (-30) &= 900 + 30 \\ &= 930 \end{aligned}$$

Therefore, removing a debt is the same as gaining a profit.

[font=12pt]

## 4.2 Subtracting Integers: KCC Rule

### Keep-Change-Change Rule

The Keep-Change-Change rule is a clever way to remember how to rewrite any subtraction problem as an addition problem.

**Procedure 6** (**Keep-Change-Change Rule**). Any subtraction problem can be rewritten as an equivalent addition problem using the steps below.

(a) **Keep** the sign of the first number.

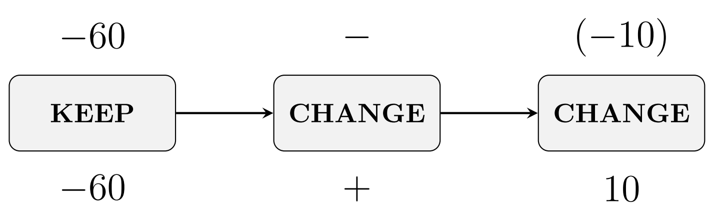
(b) **Change** the operation of subtraction to addition.

(c) **Change** the sign of the second number.

**Remember:** Subtracting a number is the same as adding the opposite of that number.

**Example 17.** Use the Keep-Change-Change rule to find the difference $-60 - (-10)$.

**Explanation.** The diagram below demonstrates how to rewrite the subtraction problem $-60 - (-10)$ as an addition problem. Starting from the left, we **keep** the sign of the first number, **change** the operation of addition to subtraction, then **change** the sign of the last number. Lastly, we add to find the result.

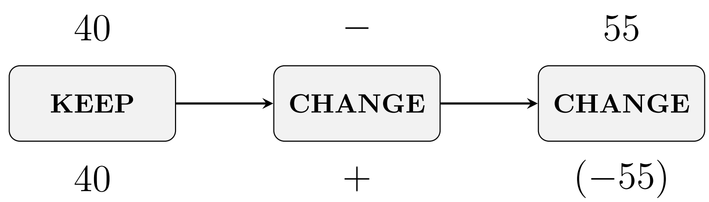



Alt text: A horizontal flowchart showing the Keep-Change-Change rule for the expression negative sixty minus negative ten. Three boxes are arranged from left to right. The first box is labeled KEEP with the number negative sixty above and below it. An arrow points to the second box, labeled CHANGE, which shows a subtraction sign above it changing to an addition sign below it. A second arrow points to the third box, labeled CHANGE, which shows negative ten above it changing to positive ten below it.

$$
\begin{aligned}
-60 - (-10) &= -60 + 10 \\
&= -50
\end{aligned}
$$

**Example 18.** Use the Keep-Change-Change rule to find the difference $40 - 55$.

**Explanation.** The diagram below demonstrates how to rewrite the subtraction problem $40 - 55$ as an addition problem. Starting from the left, we **keep** the sign of the first number, **change** the operation of addition to subtraction, then **change** the sign of the last number. Lastly, we add to find the result.




Alt text: A horizontal flowchart showing the Keep-Change-Change rule for the expression forty minus fifty five. Three boxes are arranged from left to right. The first box is labeled KEEP with the number forty above and below it. An arrow points to the second box, labeled CHANGE, which shows a subtraction sign above it changing to an addition sign below it. A second arrow points to the third box, labeled CHANGE, which shows negative fifty five above it changing to positive fifty five below it.

$$
\begin{aligned}
40 - 55 &= 40 + (-55) \\
&= -15
\end{aligned}
$$

**Problem 53** *Use the **Keep-Change-Change** rule to rewrite the subtraction problem $-8 - (-10)$ as an equivalent addition problem. Then find the answer.*

| | | |
|---|---|---|
| $-8$ | $-$ | $(-10)$ |
| ***KEEP*** | ***CHANGE*** | ***CHANGE*** |
| $-8$ | $+$ | $10$ |

***Answer*** $= 2$

**Problem 54** *Use the **Keep-Change-Change** rule to rewrite the subtraction problem $-8 - 10$ as an equivalent addition problem. Then find the answer.*

| | | |
|---|---|---|
| $-8$ | $-$ | $10$ |
| ***KEEP*** | ***CHANGE*** | ***CHANGE*** |
| $-8$ | $+$ | $-10$ |

***Answer*** $= -18$

**Problem 55** *Use the **Keep-Change-Change** rule to rewrite the subtraction problem* $200 - (-300)$ *as an equivalent addition problem.*

| | | |
|---|---|---|
| $200$ | $-$ | $(-300)$ |
| ***KEEP*** | ***CHANGE*** | ***CHANGE*** |
| $200$ | $+$ | $300$ |

***Answer*** $= 500$

**Problem 56** *Use the **Keep-Change-Change** rule to rewrite the subtraction problem* $-90 - (-20)$ *as an equivalent addition problem.*

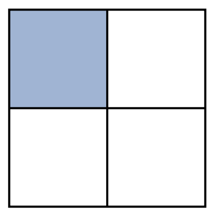
| | | |
|---|---|---|
| $-90$ | $-$ | $(-20)$ |
| ***KEEP*** | ***CHANGE*** | ***CHANGE*** |
| $-90$ | $+$ | $20$ |

***Answer*** $= -70$

# 5 Fractions: Introduction

## Introduction to Fractions

A **fraction** is an *equal* part of a whole. The square below has been divided into 4 equal parts. The fraction that represents the shaded part of the square is $\frac{1}{4}$, since 1 of the 4 parts is shaded.



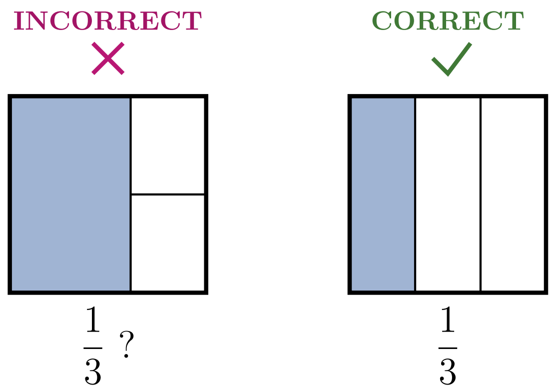
$$\frac{1}{4}$$

Alt text: A square divided into four equal quadrants by a horizontal and a vertical line through its center. The top-left quadrant is shaded blue, representing the fraction one-fourth.

The **denominator** of a fraction tells you how many equal-sized pieces a whole has been divided into. The **numerator** of a fraction is how many of those pieces you are counting or considering.

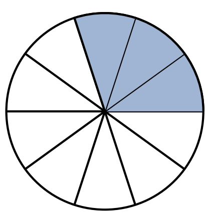
**Why do we need to divide a whole into equal parts**?

Because fractions only make sense if the whole is divided into equal parts. For example, the INCORRECT square on the left has been divided into 3 parts and 1 of them is shaded. However, we can't say that $\frac{1}{3}$ of the square is shaded because the parts are not the same size. To correct this, we redraw the lines so that every part is the exact same size, as shown in the CORRECT square on the right.




Alt text: Two squares side-by-side demonstrating the importance of equal parts in fractions. The left square is labeled INCORRECT with a red X; it is divided into three unequal parts with one large section shaded, showing that counting unequal pieces as one-third is incorrect. The right square is labeled CORRECT with a green checkmark; it is divided into three identical vertical rectangles with one shaded, correctly representing the fraction one-third.

**Problem 57** *The circle below has been divided into 10 equal parts.*

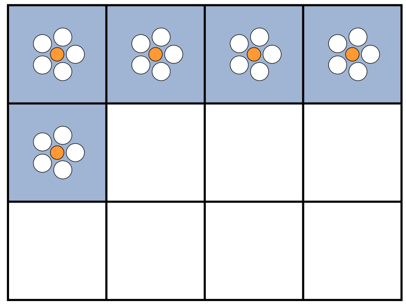


*Alt text: A circle representing a whole, divided into 10 equal pie-shaped wedges. Three adjacent wedges in the top-right section are shaded blue.*

*What fraction of the circle is shaded?*

$$\textit{Fraction Shaded} = \frac{3}{10}$$

**Problem 58** *A local wildlife sanctuary covers 12 acres of protected land. Recent data shows that 5 acres have been successfully restored with native wild flowers to support declining bee populations.*



***Each box represents 1 acre of the sanctuary***.

*Alt text: A grid of 12 equal squares arranged in three rows of four. The four squares in the top row and the first square in the middle row are shaded blue with white flowers in the center.*

***Hint:*** *The denominator in each part is the number of parts the sanctuary has been divided into. The numerator in the first part is the number of squares with flowers. The numerator in the second part is the number of squares without flowers.*

*What fraction of the sanctuary's total area has been restored with wild flowers?*

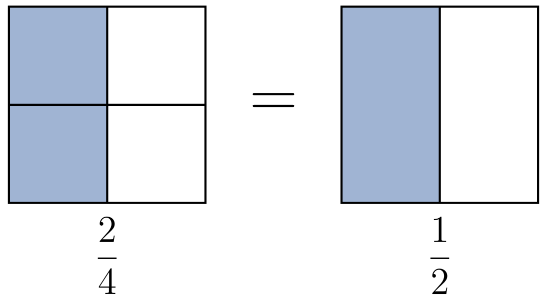
5/12

*What fraction of the sanctuary's total area has* ***not*** *been restored with wild flowers?*

7/12

## Equivalent Fractions

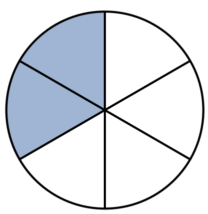
The square on the left has been divided into 4 equal parts, with 2 parts shaded. The square on the right has been divided into 2 equal parts, with 1 part shaded. Since both squares are the same size and have an equal amount of shading, we can conclude that $\frac{2}{4}$ is equivalent to $\frac{1}{2}$.




Alt text: Two squares side-by-side, as described above, separated by an equal sign. The square on the left represents two-fourths and the square on the right represents one-half.

**Equivalent fractions** represent the same part of a whole, but have different numerators and denominators. Since a whole can be divided into any number of equal parts, each fraction has infinitely many equivalent fractions.

**Problem 59** *Consider the circle below.*



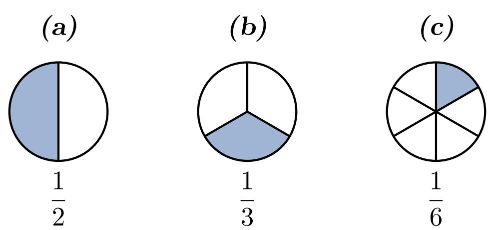
*Alt text: A circle divided into six equal wedge-shaped sections. Two adjacent sections on the upper-left are shaded blue.*

*What fraction of the circle is shaded?*

$$\textit{Fraction Shaded} = \frac{2}{6}$$

*Which of the following circles represents an **equivalent** fraction?*

((a)/ (b) ✓/ (c))




*Alt text: Three circles labeled (a), (b), and (c) representing fractions. Circle (a) is divided into 2 equal parts with 1 shaded (one-half). Circle (b) is divided into 3 equal parts with 1 shaded (one-third). Circle (c) is divided into 6 equal parts with 1 shaded (one-sixth).*

**Definition 4 (Types of Fractions).** (a) **Proper fractions** are fractions in which the numerator is smaller than the denominator. A proper fraction is always less than 1.

- **Example**: $\frac{3}{4}$

(b) **Improper fractions** are fractions in which the numerator is greater than or equal to the denominator. An improper fraction is always greater than or equal to 1.

- **Example**: $\frac{7}{12}$

(c) **Mixed numbers** consist of a whole number and a fraction. A mixed number is always greater than 1.

- **Example**: $1\frac{1}{4}$

**Problem 60** *Classify each of the following fractions as a* ***proper fraction****, an* ***improper fraction*** *or a* ***mixed number****.*

(a) $2\frac{1}{2}$

***Multiple Choice:***

(i) *Proper Fraction*

(ii) *Improper Fraction*

(iii) *Mixed Number* ✓

(b) $\frac{11}{12}$

***Multiple Choice:***

(i) *Proper Fraction* ✓

(ii) *Improper Fraction*

(iii) *Mixed Number*

(c) $\frac{15}{8}$

***Multiple Choice:***

(i) *Proper Fraction*

(ii) *Improper Fraction* ✓

(iii) *Mixed Number*

(d) $\dfrac{5}{5}$

***Multiple Choice:***

(i) *Proper Fraction*

(ii) *Improper Fraction* ✓

(iii) *Mixed Number*

**Problem 61** *When we write the mixed number* $1\frac{1}{4}$ *what arithmetic symbol is not written, but understood to be between the 1 and* $\frac{1}{4}$*?*

***Multiple Choice:***

(a) $\times$

(b) $-$

(c) $+$ ✓

(d) $\div$

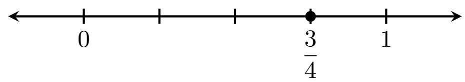
***Feedback(*`correct`*):*** *That's right!* $1\dfrac{1}{4} = 1 + \dfrac{1}{4}$

## Graphing Fractions on a Number Line

To graph a **proper fraction**, divide the distance from 0 to 1 into $n$ equal parts, where $n$ is the denominator of the fraction. Then start at 0 and use the numerator to determine the number of places to move to the right.

**Example 19.** Graph the **proper fraction** $\frac{3}{4}$.

**Explanation.** Since the denominator is 4, divide the distance from 0 to 1 into 4 equal parts. Then start at 0 and move 3 parts to the right, since the numerator is 3.

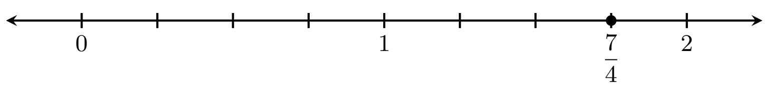



Alt text: A horizontal number line showing the interval from 0 to 1, divided into four equal segments. The integer 0 is labeled at the first tick mark, and the integer 1 is labeled at the fifth tick mark. A solid black dot is placed on the third tick mark from 0, which is labeled below with the fraction three-fourths.

To graph **improper fractions**, we continue the process and divide the distances between each whole number into $n$ equal parts.

**Example 20.** Graph the **improper fraction** $\frac{7}{4}$.

**Explanation.** Since the denominator is 4, divide the distance between each whole number into 4 equal parts. Then start at 0 and count 7 parts to the right, since the numerator is 7.



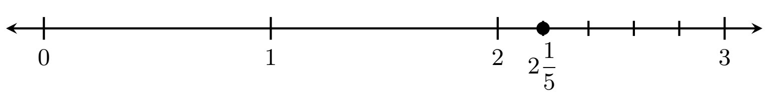

Alt text: A horizontal number line extending from 0 to 2, divided into eight equal segments. The integers 0, 1, and 2 are labeled. A single solid black dot is placed at the seventh tick mark from 0. This point is labeled below the line with the fraction seven-fourths.

**Remark 5.** Notice that $\dfrac{7}{4}$ is equivalent to $1\dfrac{3}{4}$ since it lies $\dfrac{3}{4}$ of the way between 1 and 2.

To graph a **mixed number**, first locate the whole number part on the number line. Next, divide the distance between that whole number and the next whole number into $n$ equal parts, where $n$ is the number in the denominator. Then start at the whole number part and graph the fractional part.

**Example 21.** Graph the **mixed number** $2\frac{1}{5}$.

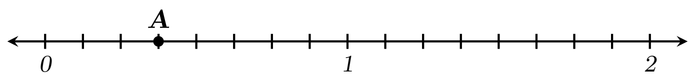
**Explanation.** Since the denominator is 5, divide the distance between the whole numbers 2 and 3 into 5 equal parts. Then start at 2 and move 1 part to the right, since the numerator is 1.




Alt text: A horizontal number line labeled with the whole numbers 0, 1, 2, and 3. The segment between 2 and 3 is divided into five equal parts. A solid black dot is placed on the first tick mark after 2, labeled as the mixed number two and one-fifth.

**Remark 6.** Negative fractions can be graphed in a similar manner.

**Problem 62** *Identify the position of the letter $A$ on the number line.*

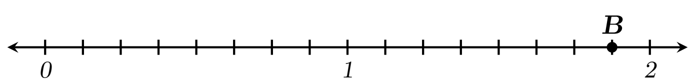



*Alt text: A horizontal number line extending from 0 to 2. The distance between each whole number is divided into eight equal parts. Point A is marked with a solid black dot and is located at the third tick mark after 0.*

$A = \frac{3}{8}$

***Hint:*** *To find the denominator of $A$, identify the number of spaces between each whole number. To find the numerator, start at 0 and count the number of tick marks to get to $A$.*

**Problem 63** *Identify the position of the letter $B$ on the number line.*




*Alt text: A horizontal number line extending from 0 to 2. The whole numbers 0, 1 and 2 are labeled. The distance between each whole number is divided into eight equal parts. Point B is located at the fifteenth tick mark after 0 or equivalently at the seventh tick mark after 1.*

***Hint:*** *To find the denominator of $B$, identify the number of spaces between each whole number. To find the numerator, start at 0 and count the number of tickmarks to get to $B$. Equivalently, start at 1 and count the number of tickmarks to get to $B$, then write the result as a mixed number.*

***Select all that apply.***

***Select All Correct Answers:***

(a) $1\frac{7}{8}$ ✓

(b) $\frac{16}{8}$

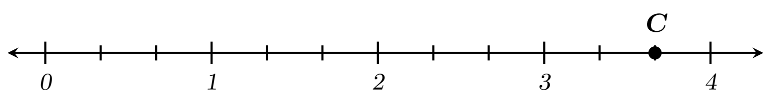
(c) $\frac{15}{8}$ ✓

(d) $1\frac{5}{8}$

(e) $\frac{5}{8}$

(f) $\frac{7}{8}$

**Problem 64** *Identify the position of the letter $C$ on the number line.*




*Alt text: A horizontal number line extending from 0 to 4. The whole numbers 0, 1, 2, 3 and 4 are labeled. The distance between each whole number has been divided into three equal parts. Point C is labeled with a solid black dot and is located at the second tick mark after 3.*

***Select all that apply*.**

***Select All Correct Answers:***

(a) $3\frac{1}{2}$

(b) $3\frac{3}{4}$

(c) $\frac{2}{3}$

(d) $\frac{11}{2}$

(e) $3\frac{2}{3}$ ✓

(f) $\frac{11}{3}$ ✓

## Negative Fractions

In the following problems, notice that the position of the negative sign changes but the value of the fraction remains the same.

$$\frac{-4}{2} = -2$$

$$\frac{4}{-2} = -2$$

$$-\frac{4}{2} = -2$$

Therefore, a negative sign in a fraction can be written in the numerator, the denominator or in front of the fraction. However, when simplifying we typically place the negative sign in front of the fraction.

[font=12pt]

## 5.1 Fractions: Prime Factorizations

### Introduction

Before we can simplify fractions, we first need to understand some concepts involving **counting numbers** (also called **natural numbers**) and **whole numbers**.

Counting Numbers: $\{1, 2, 3, \ldots\}$

Whole Numbers: $\{0, 1, 2, 3, \ldots\}$

While these sets differ only by the number 0, having distinct names is useful when discussing concepts in which it may or may not make sense to include 0. Whole numbers are used when 0 is needed to represet a starting point or the complete absence of something, while counting numbers are used when a concept requires at least one of something to exist.

**Problem 65** *Determine whether the set of* ***counting numbers*** *or the set of* ***whole numbers*** *is the most appropriate to use.*

***Rankings in a race****: Identifying the top 3 winners in a race.*

***Multiple Choice:***

(a) *Counting Numbers* ✓

(b) *Whole Numbers*

***Feedback(*****`attempt`*****):*** *There is no such thing as a 0th place in a race. The winners are 1st, 2nd and 3rd.*

**Problem 66** *Determine whether the set of* ***counting numbers*** *or the set of* ***whole numbers*** *is the most appropriate to use.*

***Points scored in a basketball game****: Tracking the total score for a team.*

***Multiple Choice:***

(a) *Counting Numbers*

(b) *Whole Numbers* ✓

***Feedback(*`attempt`*):*** *A team can be held to 0 points (a shutout).*

**Problem 67** *Determine whether the set of **counting numbers** or the set of **whole numbers** is the most appropriate to use.*

***Defining Polygons**: Stating how many sides a specific geometric shape has.*

***Multiple Choice:***

(a) *Counting Numbers* ✓

(b) *Whole Numbers*

***Feedback(*`attempt`*):*** *A shape cannot have 0 sides; it must have at least 3 sides to be a polygon.*

**Problem 68** *Determine whether the set of **counting numbers** or the set of **whole numbers** is the most appropriate to use.*

***Store Inventory**: Keeping track of how many laptops are left in stock.*

***Multiple Choice:***

(a) *Counting Numbers*

(b) *Whole Numbers* ✓

***Feedback(*`attempt`*):*** *If a store sells out of an item, the inventory count is 0.*

## Factors of Whole Numbers

The **factors of a whole number** are numbers that divide into it evenly (no remainder). Below are the factors of 18.

**Factors of 18:** 1, 2, 3, 6, 9, 18

The factors of a number can be used to write the number as a product. Below, the number 18 has been written as a product using the factors 6 and 3.

$$18 = 6 \cdot 3$$

Note that we can also write the number 18 as a product using the factors 2 and 9.

**Problem 69** *List all factors of the number 20.*

***Multiple Choice:***

(a) $1, 2, 3, 4, 5, 10, 20$

(b) $1, 2, 4, 5, 10, 20$ ✓

(c) $1, 2, 4, 5, 10, 15, 20$

(d) $1, 2, 4, 5, 10$

***Hint:*** *Write your own list of factors, rather than trying to find the answer by looking at the options. This will help you to understand and remember the process.*

**Problem 70** *List all factors of the number 24.*

***Multiple Choice:***

(a) $1, 2, 3, 4, 6, 8, 24$

(b) $1, 2, 3, 4, 6, 12, 24$

(c) $1, 2, 3, 4, 6, 8, 12, 24$ ✓

(d) $2, 3, 4, 6, 8, 12$

**Problem 71** *List all factors of the number 36.*

***Multiple Choice:***

(a) $1, 2, 3, 4, 9, 12, 18, 36$

(b) $1, 2, 3, 4, 6, 8, 9, 12, 18, 36$

(c) $1, 2, 3, 4, 6, 9, 12, 18, 36$ ✓

(d) $1, 2, 3, 4, 6, 6, 9, 12, 18, 36$

***Feedback(`attempt`):*** ***Note****: While $6 \times 6 = 36$, we only list the number 6 once when writing out the set of factors.*

## Prime and Composite Numbers

**Definition 5.** A **prime number** is a **counting number** greater than 1 that is evenly divisible *only* by itself and 1. Equivalently, a **prime number** is a **counting number** greater than 1 whose only factors are 1 and itself.

The first several prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29. A counting number greater than 1 that's not prime is called a **composite number**.

**Problem 72** *Which of the following numbers are **prime**?*

$$2, \quad 9, \quad 15, \quad 19, \quad 27, \quad 31$$

***Select all that apply.***

***Select All Correct Answers:***

(a) *2* ✓

(b) *9*

(c) *15*

(d) *19* ✓

(e) *27*

(f) *31* ✓

**Problem 73** ***True or False*** *All prime numbers are odd, except for 2.*

***Multiple Choice:***

(a) *True* ✓

(b) *False*

***Feedback(`correct`):*** *That's right! Since all even numbers are evenly divisible by 2, the only one that can be prime is 2.*

## Prime Factorization

The **prime factorization** of a composite number is the number written as the product of prime numbers. For example, the prime factorization of $18 = 2 \cdot 3 \cdot 3$. We will use **factorization trees** to find the prime factorization of a composite number.

**Procedure 7 (Finding the Prime Factorization of a Number).** To find the prime factorization of any composite number, you can create a **factorization tree** using the steps below.

**Step 1** **Find a Pair of Factors**: Identify two numbers that have a product equal to the composite number. Draw two branches extending from the composite number to the factors.

**Step 2** **Evaluate Each Factor**: If a factor is prime, circle it. This branch is now finished. If a factor is composite, do not circle it. Continue to the next step.

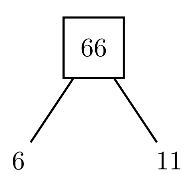
**Step 3** **Repeat the Process**: For every number that is not circled, find a pair of factors and draw new branches.

**Step 4** **Final Result**: Once every branch ends in a circled prime number, you are finished. Write the prime factorization by multiplying all the circled numbers together.

**Example 22.** Find the prime factorization of the number 66.

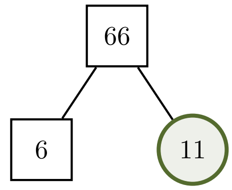
**Remark 7.** In the figures below, each composite number is written inside a square for clarity. You can omit the squares when finding prime factorizations by hand, but don't forget to circle the prime numbers!

**Explanation.** **Step 1** **Find a Pair of Factors**: The numbers 6 and 11 have a product of 66.



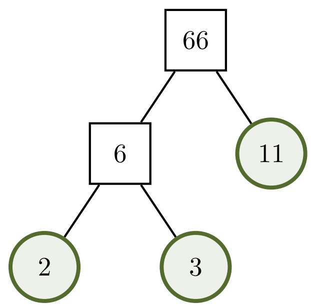

Alt text: A partial factorization tree for the number 66. The top node is 66, which branches into 6 and 11.

**Step 2** **Evaluate Each Factor**: Since 11 is prime it's circled. The number 6 is not prime, so the process is repeated in the next step.




Alt text: A partial factorization tree for the number 66. The top node is 66, which branches into 6 and 11. The number 11 is circled since it's prime.

**Step 3** **Repeat the Process**: We can write 6 as the product of 2 and 3. Since both 2 and 3 are prime, they are circled and the tree is now finished.

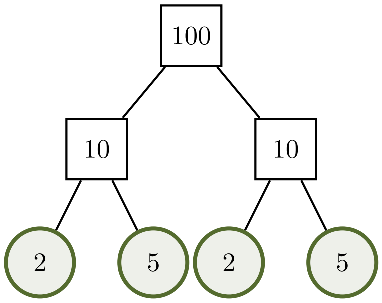



Alt text: A factorization tree for the number 66. The top node is 66, which branches into 6 and 11. The number 11 is circled since it's prime. The number 6 branches further into 2 and 3, both of which are circled since they are prime.

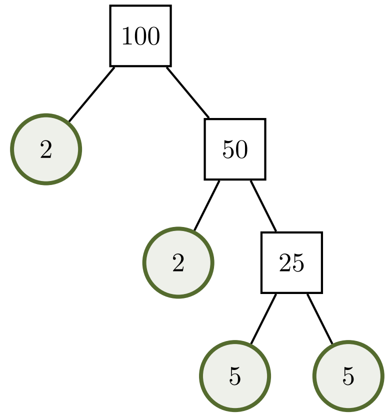
**Step 4** **Final Result**: The prime factorization of 66 is the product of all circled numbers in the tree.

$$66 = 2 \cdot 3 \cdot 11$$

**Remark 8.** No matter which factors you choose to start with, your tree will always end with the same set of prime numbers. In the figure below, there are two factorization trees for 100. One begins with the factors 10 and 10 and the other begins with 2 and 50, but they both result in the same prime factorization.



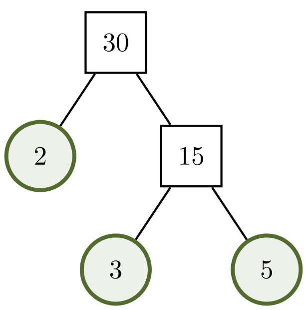

$$100 = 2 \cdot 2 \cdot 5 \cdot 5$$




$$100 = 2 \cdot 2 \cdot 5 \cdot 5$$

Alt text: Two factorization trees for the number 100 shown side-by-side. The left tree branches into two 10s, each branching into prime factors 2 and 5. The right tree branches into 2 and 50, then 50 branches into 2 and 25, then 25 branches into 5 and 5. Both trees show the final prime factorization 100 equals 2 times 2 times 5 times 5.

**Problem 74** *The factorization tree for the number 30 is given below.*




*Alt text: A factorization tree for the number 30. The top node is 30, which branches into 2 and 15. The number 2 is circled. The number 15 branches further into 3 and 5, both of which are circled.*

*Based on this tree, what is the prime factorization of* $30$*?*

***Multiple Choice:***

(a) $2 \cdot 15$

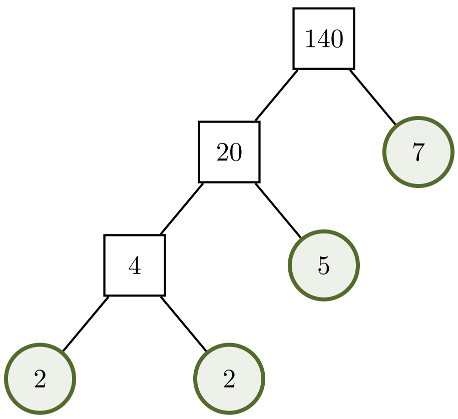
(b) $2 \cdot 3 \cdot 5$ ✓

(c) $2 \cdot 3 \cdot 10$

(d) $2 \cdot 3 \cdot 5 \cdot 15$

***Feedback(*`attempt`*):*** *To find the prime factorization, only multiply the numbers in the circles at the very ends of the branches. Composite numbers in boxes (like 15) are just stepping stones.*

**Problem 75** *The factorization tree for the number 140 is given below.*




*Alt text: A factorization tree for the number 140. The top node is 140, branching into 20 and 7. The 20 branches into 4 and 5, and the 4 branches into two circled 2s. The prime numbers 2, 2, 5, and 7 are circled.*

*Based on this tree, what is the prime factorization of* $140$*?*

***Multiple Choice:***

(a) $2 \cdot 2 \cdot 35$

(b) $4 \cdot 5 \cdot 7$

(c) $2 \cdot 2 \cdot 5 \cdot 7$ ✓

(d) $2 \cdot 5 \cdot 7 \cdot 20$

**Problem 76** *Find the prime factorization of* $42$*.*

***Multiple Choice:***

(a) $6 \cdot 7$

(b) $2 \cdot 21$

(c) $2 \cdot 3 \cdot 7$ ✓

(d) $2 \cdot 3 \cdot 3 \cdot 7$

***Hint:*** *Think of a pair of factors for* 42*. For example,* $42 = 6 \cdot 7$*. Since* 7 *is prime, you only need to factor* 6*.*

**Problem 77** *Find the prime factorization of* 120*.*

***Multiple Choice:***

(a) $10 \cdot 12$

(b) $2 \cdot 2 \cdot 3 \cdot 10$

(c) $2 \cdot 2 \cdot 2 \cdot 3 \cdot 5$ ✓

(d) $2 \cdot 3 \cdot 4 \cdot 5$

## Extra Insight: The Power of Prime Numbers

Did you know that **prime numbers** keep your online data safe from hackers? Secure messages and accounts are locked by a giant composite number and the key is its prime factorization. One way this is done is through a process called **RSA Encryption**. To create the lock, a computer picks two very large prime numbers and multiplies them together to get an enormous composite number. A computer can multiply two very large prime numbers in less than a second to create the composite number, but finding its prime factorization is very difficult! It would take the fastest supercomputer billions of years. Since its nearly impossible for hackers to find the key (the prime factorization), your accounts and messages stay locked and your information remains private!

This whole process depends on knowing two very large prime numbers. The largest known prime number was discovered in 2024 and is over 41 million digits long! People use a network of thousands of computers (called GIMPS) to hunt for these massive prime numbers nd there is even a cash prize for the person who finds the next one.

[font=12pt]

## 5.2 Fractions: GCF

### Greatest Common Factor (GCF)

In the next section, we will simplify fractions using the largest number that evenly divides into both the numerator and denominator.

**Definition 6.** The **greatest common factor** is the largest whole number that divides evenly into two or more numbers. Equivalently, it is the largest whole number that is a factor of two or more numbers.

**Problem 78** ***True or False*** *The GCF will always be less than or equal to the smallest number in the group.*

***Multiple Choice:***

(a) *True* ✓

(b) *False*

***Feedback(*`correct`*):*** *That's right! Since the GCF evenly divides into each number, it can't be larger than the smallest number.*

**Procedure 8 (Finding the GCF Using Lists of Factors).** List the factors of each number and look for the largest number that appears in each list. **Alternatively**, only list the factors of the smallest number then look for the largest factor that evenly divides into the other number(s).

**Example 23.** Find the GCF of 12 and 36.

**Explanation.** We begin by listing the factors of each number.

$$\textbf{Factors of 12}: \ 1, 2, 3, 4, 6, 12$$

$$\textbf{Factors of 36}: \ 1, 3, 4, 6, 9, 12, 18, 36$$

The *common factors* of 12 and 36 are 1, 3, 4, 6, and 12. So, the **greatest common factor** is 12.

**Remark 9.** Notice that it is more efficient to list the factors of 12 and look for the largest number that evenly divides into 36.

**Problem 79** *Find the GCF of* 20 *and* 30.

(a) *List the factors of* 20 *and* 30.

***Multiple Choice:***

(i) ***Factors of 20****: 1, 2, 4, 5, 20;* ***Factors of 30****: 1, 2, 3, 5, 6, 10, 15, 30*

(ii) ***Factors of 20****: 1, 2, 4, 5, 10, 20;* ***Factors of 30****: 1, 2, 3, 5, 10, 15, 30*

(iii) ***Factors of 20****: 1, 2, 4, 5, 10, 20;* ***Factors of 30****: 1, 2, 3, 5, 6, 10, 15, 30* ✓

(iv) ***Factors of 20****: 2, 4, 5, 10;* ***Factors of 30****: 2, 3, 5, 6, 10, 15*

(b) *Based on the lists above, what is the greatest common factor (GCF) of* 20 *and* 30*?*

$$GCF = 10$$

INSERT VIDEO SHOWING Alternative IDEA

**Problem 80** *Find the GCF of 15 and 20 by listing the factors of both numbers or just the number 15. Use whichever technique you prefer.*

$$GCF = 5$$

**Problem 81** *Find the GCF of 24 and 60 by listing the factors of both numbers or just the number 24. Use whichever technique you prefer.*

$$GCF = 12$$

**Procedure 9** (**Finding the GCF Using Prime Factorizations**)**.** Find the prime factorization of each number, then multiply the common prime factors.

**Example 24.** Find the GCF of 12 and 36 using prime factorizations.

**Explanation.** We begin by finding the prime factorization of each number.

$$\begin{aligned} 12 &= 2 \cdot 2 \cdot 2 \cdot 3 \\ 36 &= 2 \cdot 2 \cdot 3 \cdot 3 \end{aligned}$$

The common factors are 2 and 3. There are two factors of 2 and one factor of 3 in *both* factorizations, therefore

$$\begin{aligned} \text{GCF} &= 2 \cdot 2 \cdot 3 \\ &= 12 \end{aligned}$$

**Problem 82** *Find the GCF of 60 and 75 using their prime factorizations given below.*

$$\begin{aligned} 60 &= 2 \cdot 2 \cdot 3 \cdot 5 \\ 75 &= 3 \cdot 5 \cdot 3 \end{aligned}$$

$GCF = 15$

***Feedback(`attempt`):*** *Excellent! The common factors are 3 and 5. There is one factor of 3 and one factor of 5 in **both** factorizations, so the GCF $= 3 \cdot 5$ which equals 15.*

**Problem 83** *Find the GCF of 85 and 150 using their prime factorizations shown below.*

$$\begin{aligned} 85 &= 5 \cdot 17 \\ 150 &= 2 \cdot 3 \cdot 5 \cdot 5 \end{aligned}$$

$GCF = 5$

INSERT VIDEO: Finding the GCF of 3 numbers!!!!!

The GCF can be used to solve problems involving groups of things that need to be divided efficiently, with no waste. Use what you've learned about finding the GCF to solve the following application problems.

**Problem 84** *You are creating identical goodie bags for a child's birthday party. You have* ***18*** *Airheads,* ***24*** *Tootsie Rolls and* ***36*** *Kit-Kats. What is the* ***greatest*** *number of identical goodie bags you can make if you want to use all of the candy?*

$$\text{Greatest Number of Goodie Bags} = 6$$

***Hint:*** *To solve this, you need to find the Greatest Common Factor (GCF) of 18, 24, and 36. List the factors for each number and find the largest one that appears in all three lists.*

*Now that you know you can make identical goodie bags with no waste, how many of each candy will be in* ***each*** *bag?*

- *Airheads:* 3
- *Tootsie Rolls:* 4
- *Kit-Kats:* 6

***Hint:*** *Divide the number of each candy by the number of bags. For example,* $18 \div 6 = 3$ *Airheads.*

**Problem 85** *A carpenter has three wooden boards of different lengths:* 8 *ft,* 12 *ft and* 20 *ft. The carpenter wants to cut all three boards into smaller pieces of equal length, so that there is no scrap wood left over. What is the longest possible length for these pieces?*

$$\text{Longest Length} = 4 \text{ feet}$$

***Hint:*** *Find the GCF of 8, 12, and 20. This is largest number that divides evenly into all three lengths and is therefore the longest length.*

*How many pieces will the carpenter have after cutting all three boards?*

$$\text{Total Pieces} = 10$$

***Hint:*** *Divide each board's length by the GCF to see how many pieces come from each board, then add them.*

[font=12pt]

## 5.3 Fractions: Simplifying

### Simplifying Fractions

A fraction is said to be in **simplest form** or **lowest terms** when the numerator and denominator have no common factors other than 1. For example, the fraction $\frac{2}{5}$ is in simplest form because 2 and 5 have no common factors. However, the fraction $\frac{4}{6}$ is not in simplest form because both 4 and 6 have a common factor of 2. The process of writing a fraction in simplest form is called **simplifying** the fraction. We will consider two methods to simplify fractions. The first method uses the GCF and the second method uses prime factorizations.

**Procedure 10 (Simplifying Fractions Using the GCF).** Divide the numerator and denominator by the GCF.

**Example 25.** Simplify the fraction $\dfrac{4}{12}$ using the GCF $= 4$.

**Explanation.** Divide the numerator and denominator by 4 and write the result.

$$\begin{aligned}\frac{4}{12} &= \frac{4 \div 4}{12 \div 4}\\ &= \frac{1}{3}\end{aligned}$$

Done!

We can also simplify fractions by dividing the numerator and denominator by *any* common factor, however using the GCF involves just one step. If we didn't notice that the GCF was 4, but knew that both numbers were evenly divisible by 2, we would need two steps to simplify the fraction $\frac{4}{12}$. First, we divide the numerator and denominator by 2, then we divide by 2 again.

$$\begin{aligned}\frac{4}{12} &= \frac{4 \div 2}{12 \div 2}\\ &= \frac{2}{6}\\ &= \frac{2 \div 2}{6 \div 2}\\ &= \frac{1}{3}\end{aligned}$$

**Problem 86** *Simplify the fraction* $\frac{24}{36}$ *using the* $GCF = 12$.

$$\frac{24}{36} = \frac{2}{3}$$

**Problem 87** *Follow the steps below to simplify the fraction* $\frac{18}{45}$.

(a) *Find the Greatest Common Factor (GCF) of 18 and 45.*

$$GCF = 9$$

(b) *Divide both the numerator and the denominator by the GCF to simplify.*

$$\frac{18}{45} = \frac{2}{5}$$

***Hint:*** *List the factors for each number to find the largest one they share (GCF):*

- *Factors of 18: 1, 2, 3, 6, 9, 18*
- *Factors of 45: 1, 3, 5, 9, 15, 45*

**Procedure 11** (**Simplifying Fractions Using Prime Factorizations**). (a) Write the prime factorization of the numbers in the numerator and denominator.

(b) Cancel (divide out) the common factors that appear in both the the numerator and denominator.

(c) Multiply the remaining numbers in the numerator and denominator to find the result.

**Remark 10.** To **cancel** or divide out means that we remove the common factors, since their quotient is 1.

**Example 26.** Simplify the fraction $\frac{20}{25}$ using prime factorizations.

**Explanation.** Begin by writing the prime factorizations of 20 in the numerator and 25 in the denominator. Since there is a common factor of 5 in the numerator and denominator, we can cancel or divide out the fives since $\frac{5}{5} = 1$. Finally, we multiply the remaining numbers in the numerator and denominator to find the result.

$$\begin{aligned}\frac{20}{25} &= \frac{2 \cdot 2 \cdot 5}{5 \cdot 5} \\ &= \frac{2 \cdot 2 \cdot \cancel{5}}{5 \cdot \cancel{5}} \\ &= \frac{4}{5}\end{aligned}$$

INSERT VIDEO of simplifying a fraction using one of these facts.

**Problem 88** *Simplify the fraction using the prime factorizations shown below.*

$$\begin{aligned} 42 &= 2 \cdot 3 \cdot 7 \\ 70 &= 2 \cdot 5 \cdot 7 \end{aligned}$$

$\frac{42}{70} = \frac{3}{5}$

***Hint:*** *Write the prime factorizations in the fraction, then cancel the common prime numbers.*

$$\frac{42}{70} = \frac{2 \cdot 3 \cdot 7}{2 \cdot 5 \cdot 7}$$

**Problem 89** *Simplify the fraction using whichever method you prefer.*

$$\frac{33}{60} = \frac{11}{20}$$

***Hint:*** *If you want to use prime factorizations, write the prime factorizations in the fraction, then cancel the common prime numbers.*

$$\frac{33}{60} = \frac{3 \cdot 11}{2 \cdot 2 \cdot 3 \cdot 5}$$

***Hint:*** *If you want to use the Greatest Common Factor (GCF) to simplify, the factors of 33 are $1, 3, 11, 33$. The largest factor that evenly divides into 60 is the GCF.*

**Problem 90** *Simplify the fraction using whichever method you prefer.*

$$\frac{60}{126} = \frac{10}{21}$$

***Hint:*** *If you want to use prime factorizations, write the prime factorizations in the fraction, then cancel the common prime numbers.*

$$\frac{60}{126} = \frac{2 \cdot 2 \cdot 3 \cdot 5}{2 \cdot 3 \cdot 3 \cdot 7}$$

***Hint:*** *If you want to use the Greatest Common Factor (GCF) to simplify, the factors of* $60$ *are* $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$. *The largest factor that evenly divides into* $126$ *is the GCF.*

**Problem 91** *A recycling center collected* $54$ *tons of plastic last month. However, due to contamination (like food waste or non-recyclable items), only* $21$ *tons could be recycled. What* ***fraction*** *of the collected plastic was recycled?* ***Remember to simplify your answer, if possible****.*

$$\textit{Fraction Recycled} = \frac{7}{18}$$

***Hint:*** *Write the fraction as the amount recycled over the total amount collected, then simplify.*

$$\frac{21}{54}$$

**Problem 92** *An animal rescue organization saved* $120$ *dogs from an overcrowded shelter. Of these dogs,* $45$ *were seniors who required specialized medical care and longer stays before they were ready for adoption. What* ***fraction*** *of the rescued dogs were seniors?* ***Remember to simplify your answer, if possible****.*

$$\textit{Fraction of Senior Dogs} = \frac{3}{8}$$

***Hint:*** *Write the fraction as the number of senior dogs over the total number of dogs rescued, then simplify.*

$$\frac{45}{120}$$

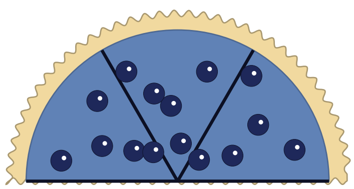
[font=12pt]

## 5.4 Fractions: Multiplication

### Multiplying Fractions

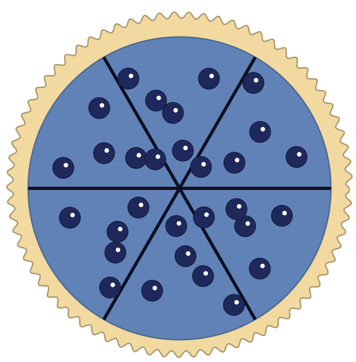
If you have half of a blueberry pie left in the fridge and eat $\frac{2}{3}$ **of** it, what fraction of the **entire pie** did you eat?

To eat $\frac{2}{3}$ **of** the half, you would need to divide the half into 3 equal pieces and eat 2 of them.



Alt text: A semi-circle representing the top half of a blueberry pie that is cut into three equal pieces.

If the **whole** pie was divided into slices of that same size, there would be a total of 6 slices.



Alt text: A circle representing whole blueberry pie that is cut into six equal pieces.

Since you ate 2 slices, you ate $\frac{2}{6}$ or $\dfrac{1}{3}$ of the **whole** pie.

In mathematics, the word **of** is code for multiplication. Therefore, to find $\frac{2}{3}$ **of** $\frac{1}{2}$, we multiply the fractions. To obtain the answer $\frac{2}{6}$ mathematically, we would have to multiply the numerators and the denominators and this is exactly how we multiply fractions.

$$\frac{2}{3} \textbf{ of } \frac{1}{2} = \frac{2}{3} \cdot \frac{1}{2}$$

$$= \frac{2}{6}$$

$$= \frac{1}{3}$$

**Procedure 12** (Multiplying Fractions)**.** If $a, b, c$ and $d$ are numbers such that $b$ and $d$ are not equal to 0, then

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$$

To multiply fractions, multiply the numerators and the denominators.

**Remark 11.** (a) It's a good idea to simplify fractions before multiplying. If you do so, you won't have to simplify the answer.

(b) Mixed numbers must be written as improper fractions before multiplying.

(c) When you multiply a fraction and a whole number, it can be helpful to write the whole number as a fraction with a denominator of 1. For example, if you want to find $\frac{1}{8}$ **of** 15, you can write 15 divided by 1 as shown below.

$$\frac{1}{8} \cdot \frac{15}{1}$$

Now, it's clear which numbers are in the numerator and denominator of each fraction.

**Problem 93** *Find each product. Remember to simplify your answer, if possible.*

$$\frac{3}{5} \times \frac{4}{7} = \frac{12}{35}$$

$$\frac{12}{8} \times \frac{1}{4} = \frac{3}{8}$$

$$50 \times \frac{25}{100} = \frac{25}{2}$$

**Problem 94** *A cookie recipe calls for* $\dfrac{3}{4}$ *cup of flour. If you only want to make* $\dfrac{2}{3}$ *of the recipe, how many cups of flour do you need?*

$$\frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2} \textit{ cups}$$

**Problem 95** *Financial advisors recommend spending no more than* $\frac{3}{10}$ *of your monthly take-home pay (your income after taxes and health insurance) on housing costs, like rent or a mortgage. If your monthly take-home pay is $2500, what is the maximum amount you should pay for rent or a mortgage?*

$$\frac{3}{10} \cdot 2500 = \$\ 750$$

***Hint:*** *Start by writing 2500 as* $\dfrac{2500}{1}$*, then simplify before multiplying. One way is to divide 2500 by 10, then multiply the result by 3 to get the answer.*

[font=12pt]

## 5.5 Fractions: Division

### Dividing Fractions

**Definition 7** (**Reciprocal of a Fraction**)**.** The **reciprocal** of the fraction $\frac{a}{b}$ is $\frac{b}{a}$.

**Remark 12.** (a) The reciprocal of a fraction is the fraction with the numerator and denominator interchanged (inverted).

(b) The **reciprocal of an integer** is found by writing the integer over 1 and then interchanging the numerator and denominator. For example, to find the reciprocal of $-4$, write it as $-\frac{4}{1}$, then interchange the numerator and denominator to get $-\frac{1}{4}$.

(c) Two numbers are reciprocals of each other if their product is 1.

$$\frac{a}{b} \cdot \frac{b}{a} = 1$$

(d) Every number has a reciprocal except 0. This is because if we write 0 as $\frac{0}{1}$ and then take the reciprocal we get $\frac{1}{0}$ which is undefined. Furthermore, if 0 had a reciprocal then the product of 0 and its reciprocal should equal 1, but we know that the product of 0 and any number equals 0.

**Problem 96** *Find the reciprocal of* $\frac{6}{11}$.

$$\frac{11}{6}$$

**Problem 97** *Find the reciprocal of* 18.

$$\frac{1}{18}$$

**Definition 8 (Dividing Fractions).** If $a, b, c$ and $d$ are numbers and $b, c$ and $d$ are not 0, then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

To divide with fractions, multiply the first fraction by the reciprocal of the second fraction.

**Warning 2.** DO NOT SIMPLIFY until you have written the division problem as an equivalent multiplication problem. In the division problem below, you might think that the fives cancel, however by rewriting it as multiplication, you can see that they do not.

$$\frac{2}{5} \div \frac{5}{3} = \frac{2}{5} \cdot \frac{3}{5}$$

**Problem 98** *Divide. Remember to simplify your answer, if possible.*

$$\frac{10}{11} \div \frac{4}{5} = \frac{25}{22}$$

$$7 \div \frac{42}{3} = \frac{1}{2}$$

$$\frac{9}{8} \div \frac{3}{7} = \frac{21}{8}$$

**Problem 99** *You have a ribbon that is $\frac{5}{6}$ of a yard long. You need to cut the ribbon into shorter pieces, each measuring exactly $\frac{1}{12}$ of a yard. How many pieces can you cut from the ribbon? Begin by setting up the division problem, then find the answer.*

***Multiple Choice:***

(a) $\frac{5}{6} \div 12$

(b) $\frac{5}{6} \div \frac{1}{12}$ ✓

(c) $\frac{1}{12} \div \frac{5}{6}$

(d) $12 \div \frac{5}{6}$

10 *pieces*

***Hint:*** *Divide the total length by the length of each piece.*

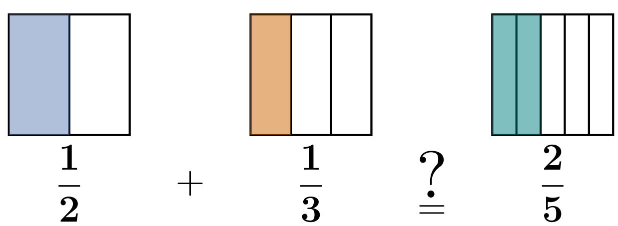
## 5.6 Fractions: Adding and Subtracting

### A Common Mistake

A student is asked to add $\frac{1}{2}$ and $\frac{1}{3}$ and writes the following. Is their work correct?

$$\frac{1}{2}+\frac{1}{3}=\frac{2}{5}$$

To answer this question, let's look at a visual representation of the student's logic.




Alt text: Three squares are shown. The first is divided into 2 equal vertical parts with 1 shaded, labeled one-half. A plus sign appears. The second is divided into 3 equal vertical parts with 1 shaded, labeled one-third. An equal sign with a question mark above it appears. The third square is divided into 5 equal vertical parts with 2 shaded, labeled two-fifths.

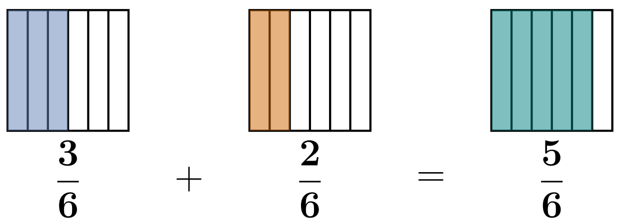
Since $\frac{1}{3}$ is *added* to $\frac{1}{2}$, the result should be greater than $\frac{1}{2}$. However, you can see that the student's answer of $\frac{2}{5}$ is actually less than $\frac{1}{2}$. This shows that simply adding the numerators and denominators doesn't work! To get the correct answer, we need to divide each whole into the same number of pieces.

**Fractions can only be added or subtracted when the pieces are the same size.**

Recall that the denominator of a fraction tells you how many equal-sized pieces a whole has been divided into. If we decide to divide each whole into six equal pieces, the original fractions can be rewritten as equivalent fractions with a denominator of 6 by multiplying by 1 in the appropriate form. Below, the fraction $\dfrac{1}{2}$ has been multiplied by 1 in form $\dfrac{3}{3}$ and the fraction $\dfrac{1}{3}$ has been multiplied by 1 in the form $\dfrac{2}{2}$.

$$\begin{aligned}\frac{1}{2} &= \frac{1}{2}\cdot\frac{3}{3} \\ &= \frac{3}{6}\end{aligned} \qquad\qquad \begin{aligned}\frac{1}{3} &= \frac{1}{3}\cdot\frac{2}{2} \\ &= \frac{2}{6}\end{aligned}$$

Now we can visually represent the sum $\frac{1}{2}+\frac{1}{3}$ as an equivalent sum with equal pieces:



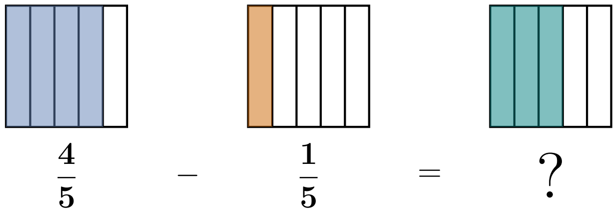

Alt text: Three rectangles are shown, each divided into 6 equal vertical parts. The first has 3 parts shaded, labeled three-sixths. A plus sign appears. The second has 2 parts shaded, labeled two-sixths. An equals sign appears. The third has 5 parts shaded, labeled five-sixths, showing the correct sum.

Combining 3 sixths and 2 sixths results in a total of 5 sixths. The entire problem is written mathematically below:

$$\begin{aligned}\frac{1}{2}+\frac{1}{3} &= \frac{1}{2}\cdot\frac{3}{3}+\frac{1}{3}\cdot\frac{2}{2} \\ &= \frac{3}{6}+\frac{2}{6} \\ &= \frac{5}{6}\end{aligned}$$

**Remark 13.** To learn mathematics, your solutions should be written using clear, logical steps as shown above; this process will deepen your understanding, prevent errors and serve as a test of your knowledge—if you cannot clearly communicate your work, you have not yet fully grasped the concept.

**Problem 100** *Use the visual representation to find the difference.*




*Alt text: A visual fraction subtraction problem shown using three identical rectangles divided vertically into five equal columns. The first rectangle has four columns shaded and is labeled four-fifths. This is followed by a minus sign. The second rectangle has one column shaded and is labeled one-fifth. This is followed by an equal sign. The third rectangle has three columns shaded and is labeled with a large question mark.*

$\frac{4}{5} - \frac{1}{5} = \frac{3}{5}$

**Procedure 13** (**Adding and Subtracting Fractions**). **Step1: Find a Common Denominator** Before adding or subtracting, the pieces must be the same size. If the denominators are different, rewrite the fractions as equivalent fractions with a common denominator.

**Step 2: Add or Subtract the Numerators** This tells you how many pieces you have.

**Steep 3: Keep the Denominator the Same** The size of the pieces doesn't change during addition or subtraction, so the denominator stays the same.

**Step 4: Simplify** Check to see if the resulting fraction can be simplified. If so, simplify.

## The Least Common Denominator

When adding or subtracting fractions with different denominators, we will use the *smallest* or *least common denominator*. This will make the simplification at the end as easy as possible.

**Definition 9** (**Multiples**). The **multiples of a number** are found by multiplying the number by the counting numbers: $\{1, 2, 3, 4, \ldots\}$. For example, the first three multiples of 4 are: 4, 8 and 12.

**Problem 101** *List the first five multiples of* 3.

***Multiple Choice:***

(a) *0, 3, 6, 9, 12*

(b) *3, 6, 9, 12, 15* ✓

(c) *6, 9, 12, 15, 18*

(d) *3, 9, 12, 15, 18*

**Definition 10** (**Least Common Denominator LCD**)**.** The **least common denominator** is the smallest number that is a multiple of two or more denominators. Equivalently, it is the smallest number that is evenly divisible by two or more denominators.

**Procedure 14** (**Finding the Least Common Denominator**)**.** Write the multiples of each denominator until you find the smallest one that appears in each list. **Alternatively**, you can list the multiples of only the largest denominator and stop when you find the smallest one that is evenly divisible by the other denominators.

**Example 27.** Find the LCD of $\dfrac{3}{4}$ and $\frac{1}{7}$.

**Explanation.** List the multiples of 4 and 7 as shown below.

$$\begin{aligned}&\textbf{Multiples of 4}: 4, 8, 12, 16, 20, 24, \mathbf{28}, 32, 36, \ldots\\&\textbf{Multiples of 7}: 7, 14, 21, \mathbf{28}, 35, 42, 49, 56, 63, \ldots\end{aligned}$$

Since 28 is the smallest number that appears in *both* lists, the LCD = 28.

**Remark 14.** Note that it is more efficient to list only the multiples of 7 and look for the smallest number in the list that is evenly divisible by 4. Had we done so, the list would've stopped at 28.

**Problem 102** *Now that you know that the LCD of $\dfrac{3}{4}$ and $\frac{1}{7}$ is 28, rewrite each fraction as an equivalent fraction with a denominator of 28. Then add the fractions and simplify, if possible.*

$$\frac{3}{4} = \frac{21}{28}$$

$$\frac{1}{7} = \frac{4}{28}$$

***Hint:*** *Mutliply each fraction by 1 in the appropriate form. For example, to write $\dfrac{3}{4}$ as an equivalent fraction with a denominator of 28, multiply it by 1 in the form $\dfrac{7}{7}$.*

$\dfrac{3}{4} + \dfrac{1}{7} = \dfrac{25}{28}$

**Problem 103** *Follow the steps below to find the sum* $\frac{1}{4}+\frac{2}{3}$.

(a) *Find the LCD of the fractions.*

$$LCD = 12$$

***Hint:*** *To find the LCD, list the multiples of 4 and look for the smallest one that is evenly divisible by 3.*

(b) *Rewrite each fraction as an equivalent fraction with the least common denominator.*

$$\frac{1}{4} = \frac{3}{12}$$

$$\frac{2}{3} = \frac{8}{12}$$

(c) *Add the fractions. Simplify your answer, if possible.*

$$\frac{1}{4}+\frac{2}{3} = \frac{11}{12}$$

**Problem 104** *Follow the steps below to find the difference* $\frac{7}{10}-\frac{1}{6}$.

(a) *Find the LCD of the fractions.*

$$LCD = 30$$

(b) *Rewrite each fraction as an equivalent fraction with the least common denominator.*

$$\frac{7}{10} = \frac{21}{30}$$

$$\frac{1}{6} = \frac{5}{30}$$

(c) *Subtract the fractions. Simplify your answer, if possible.*

$$\frac{7}{10}-\frac{1}{6} = \frac{8}{15}$$

**Problem 105** *A community land trust is dividing a large lot to combat food insecurity. They decide to use $\frac{2}{3}$ of the lot for growing fresh vegetables to donate to local food pantries and $\frac{1}{5}$ of the lot for a fruit orchard. Follow the steps below to find the fraction of the land that is dedicated to food production.*

(a) *Find the LCD of the fractions.*

$$LCD = 15$$

***Hint:*** *To find the LCD, list the multiples of 5 and look for the smallest one that is evenly divisible by 3.*

(b) *Rewrite each fraction as an equivalent fraction with the least common denominator.*

$$\text{Vegetable Garden: } \frac{2}{3} = \frac{10}{15}$$

$$\text{Fruit Orchard: } \frac{1}{5} = \frac{3}{15}$$

(c) *Add the fractions. Simplify your answer, if possible.*

$$\text{Fraction of Land Used for Food } = \frac{13}{15}$$

**Problem 106** *Factory farms store animal waste in massive open-air pits called lagoons. In one community, a lagoon is $\frac{5}{6}$ full of untreated waste. To lower the level and prevent an overflow into the local groundwater, the farm operators spray an amount equal to $\frac{1}{4}$ of the lagoon's total capacity into the air and onto nearby fields. Follow the steps below to determine the fraction of the lagoon's total capacity that remains full.*

(a) *Find the LCD of the fractions.*

$$LCD = 12$$

***Hint:*** *To find the LCD, write the multiples of 6 and look for the smallest number that is evenly divisible by 4. Or write the multiples of both numbers and look for the smallest number that appears in both lists.*

(b) *Rewrite each fraction as an equivalent fraction with the least common denominator.*

$$\textit{Initial Waste Level: } \frac{5}{6} = \frac{10}{12}$$

$$\textit{Waste Removed: } \frac{1}{4} = \frac{3}{12}$$

(c) *Subtract the fractions. Simplify your answer, if possible.*

$$\textit{Remaining Waste Level} = \frac{7}{12}$$

***Feedback(`attempt`):*** *While spraying the waste on fields prevents the lagoon from overflowing, it creates a new problem for the community. The waste is often sprayed as a fine mist that can travel for miles, carrying harmful bacteria, nitrates and intense odors into nearby neighborhoods. For the people living nearby, this isn't just an unpleasant smell—it is a serious health hazard! Studies have shown that families living near these spray fields experience higher rates of asthma, respiratory infections and even contaminated well water.*